1 Osnove teorije

1.1 BCS teorija pri končni temperaturi

BCS model, kjer je $\xi_{\mathbf{k},\sigma} = \epsilon_{\mathbf{k},\sigma} - E_F$ in $k := (\mathbf{k}, \uparrow)$, $-k := (-\mathbf{k}, \downarrow)$:

$$H = \sum_k \xi_k (c_k^\dagger c_k + c_{-k}^\dagger c_{-k}) - V \sum_{k,k'} c_k^\dagger c_{-k}^\dagger c_{-k'} c_{k'} \,. \tag{1}$$

Naredimo približek povprečnega polja, definiramo $b_k = c_k c_{-k}$ in $b_k = \langle b_k \rangle + \delta b_k = F_k + \delta b_k$, sledi (do $\mathcal{O}(\delta b_k^2)$)

$$b_k^\dagger b_{k'} \simeq F_k^* F_{k'} + F_k^* \delta b_{k'} + F_{k'} \delta b_k^\dagger = F_k^* b_{k'} + F_{k'} b_k^\dagger - F_k^* F_{k'} = \langle b_k^\dagger \rangle b_{k'} + b_k^\dagger \langle b_{k'} \rangle - \langle b_k^\dagger \rangle \langle b_{k'} \rangle \,. \tag{2}$$

Iz česar sledi:

$$H = \sum_k \begin{pmatrix} c_k^\dagger & c_{-k} \end{pmatrix} \begin{pmatrix} \xi_k & -\Delta_k \\ -\Delta_k^* & -\xi_k \end{pmatrix} \begin{pmatrix} c_k \\ c_{-k}^\dagger \end{pmatrix} + \frac{|\Delta|^2}{V} + \sum_k \xi_k \,. \tag{3}$$

Ob definiciji $\Delta_k = \sum_{k'} V(k-k') \langle c_{k'} c_{-k'} \rangle$, vzamemo še lokalno interakcijo, zato je $V(q) \simeq V$ in $\Delta_k \simeq \Delta$. Naredimo Bogoljubov rotacijo, privzamemo simetrijo okoli E_F: $u_k = u_{-k}$ ter $v_k = v_{-k}$. Iz CCR sledi $u_k^2 + v_k^2 = 1$. Kvazidelci so definirani z rotacijo

$$\begin{pmatrix} c_k \\ c_{-k}^\dagger \end{pmatrix} = \begin{pmatrix} u_k & v_k \\ -v_k & u_k \end{pmatrix} \begin{pmatrix} \alpha_k \\ \alpha_{-k}^\dagger \end{pmatrix} \,. \tag{4}$$

Nov Hamiltonjan pa postane

$$H = \sum_k \begin{pmatrix} \alpha_k^\dagger & \alpha_{-k} \end{pmatrix} \begin{pmatrix} \xi_k u_k^2 + u_k v_k (\Delta_k + \Delta_k^*) - \xi_k v_k^2 & -2\xi_k u_k v_k + \Delta_k u_k^2 - \Delta_k^* v_k^2 \\ -2\xi_k u_k v_k - \Delta_k v_k^2 + \Delta_k^* u_k^2 & \xi_k v_k^2 - \xi_k u_k^2 + u_k v_k (\Delta_k + \Delta_k^*) \end{pmatrix} \begin{pmatrix} \alpha_k \\ \alpha_{-k}\,, ^\dagger \end{pmatrix} \,, \tag{5}$$

kjer zahtevamo, da so izvendiagonalni členi enaki 0. Iz CCR zahteve lahko vzamemo parameterizacijo (tu sem vzel kompleksno nastavimo, da je $\Delta = \Delta^* e^{i2\phi_k}$):

$$v_k = \sin\left(\frac{\theta_k}{2}\right) e^{i\phi_k} \,, \tag{6}$$

$$u_k = \cos\left(\frac{\theta_k}{2}\right) \,, \tag{7}$$

$$u_k v_k = \cos\left(\frac{\theta_k}{2}\right) \sin\left(\frac{\theta_k}{2}\right) = \frac{1}{2} \sin\theta_k \,, \tag{8}$$

$$u_k^2 - v_k^2 = \cos^2\left(\frac{\theta_k}{2}\right) - \sin^2\left(\frac{\theta_k}{2}\right) = \cos\theta_k \,, \tag{9}$$

skupaj dobimo (od tu naprej $\Delta := |\Delta|$):

$$\xi_k \sin(\theta_k) = \Delta \cos(\theta_k) \,. \tag{10}$$

kar fiksira

$$\tan(\theta_k) = \frac{|\Delta_k|}{\xi_k} \,. \tag{11}$$

Tako iz trikotnika preberemo

$$\sin\theta_k = \frac{|\Delta_k|}{E_k} = \frac{|\Delta_k|}{\sqrt{\xi_k^2 + |\Delta_k|^2}} \,, \tag{12}$$

$$\cos\theta_k = \frac{\xi_k}{E_k} = \frac{\xi_k}{\sqrt{\xi_k^2 + |\Delta_k|^2}} \,, \tag{13}$$

$$\tag{14}$$

kjer je $E_k = \pm\sqrt{\xi_k^2 + |\Delta_k|^2}$ energija kvazidelcev:

$$H = \sum_k E_k \alpha_k^\dagger \alpha_k - E_k \alpha_{-k}^\dagger \alpha_{-k} + \xi_k + \frac{|\Delta|^2}{V} \,. \tag{15}$$

V termičnem stanju so diagonalni delci porazdeljeni po

$$\langle \alpha_k^\dagger \alpha_k \rangle = \frac{1}{1+\exp(E_k/T)} = \frac{1}{2} - \frac{1}{2}\tanh\left(\frac{E_k}{2T}\right) \,. \tag{16}$$

Uporabni sta zvezi $f(-E/T) = 1 - f(E/T)$ in $1 - 2f(E/T) = \tanh(E/(2T))$. V definiciji ureditvenega parametra je

$$\langle c_k c_{-k} \rangle = \left\langle \left(u_k \alpha_k - v_k \alpha_{-k}^\dagger\right)\left(u_k \alpha_{-k} + v_k \alpha_k^\dagger\right) \right\rangle = u_k v_k \left(1 - \langle \alpha_{-k}^\dagger \alpha_{-k}\rangle - \langle \alpha_k^\dagger \alpha_k \rangle\right) = \frac{1}{2}\sin\theta_k \tanh\left(\frac{\beta E_k}{2}\right) \,, \tag{17}$$

iz česar sledi enačba za režo pri $T > 0$ (seštevamo po k okoli E_F, zato $'$ nad vsoto):

$$\Delta = \frac{V}{2}{\sum_k}' \sin\theta_k \tanh\left(\frac{E_k}{2T}\right) = \frac{V\Delta}{2}\sum_k \frac{\tanh\left(\frac{1}{2T}\sqrt{\Delta^2+\xi_k^2}\right)}{\sqrt{\Delta^2+\xi_k^2}} \,. \tag{18}$$

1.2 Minimalni model ekscitonskega izolatorja pri končni temperaturi

Pri izolatorju za minimalni model vzamemo dve orbitali (a, b) model tesne vezi v vsaki posebej in lokalno Coulombovo interakcijo na mestu j med elektronskima gostotama v posameznih pasovih

$$H = H_0 + H_V = \sum_{i,j}\left(\epsilon_a^{(0)}\delta_{ij} + t_{ij}^{(a)}\right) c_{a,i}^\dagger c_{a,j} + \sum_{i,j}\left(\epsilon_b^{(0)}\delta_{ij} + t_{ij}^{(b)}\right) c_{b,i}^\dagger c_{b,j} + V\sum_j c_{a,j}^\dagger c_{a,j} c_{b,j}^\dagger c_{b,j} \tag{19}$$

$$= \sum_{\mathbf{k}} \varepsilon_{a,\mathbf{k}} c_{a,\mathbf{k}}^\dagger c_{a,\mathbf{k}} + \sum_{\mathbf{k}} \varepsilon_{b,\mathbf{k}} c_{b,\mathbf{k}}^\dagger c_{b,\mathbf{k}} + V\sum_j c_{a,j}^\dagger c_{a,j} c_{b,j}^\dagger c_{b,j} \tag{20}$$

najprej diagonaliziramo vsako orbitalo posebej, kar da gole disperzije (vezavni energiji nastavimo na $\pm G/2$) (delamo v dveh dimenzijah in mrežna konstanta $a_0 = 1$):

$$\varepsilon_{a(b),\mathbf{k}} = \pm\frac{G}{2} - 2t_{a(b)}(\cos(k_x) + \cos(k_y)) \,, \tag{21}$$

v tem primeru je najmanjša razdalja med pasovoma $G_0 = G - 4(t_a - t_b)$ oziroma $G_0 = G - 8t_a$, če je $t_b = -t_a$. Za interakcijo spet naredimo približek povprečnega polja:

$$c_{a,j}^\dagger c_{a,j} c_{b,j}^\dagger c_{b,j} \approx \langle c_{a,j}^\dagger c_{a,j}\rangle c_{b,j}^\dagger c_{b,j} + \langle c_{b,j}^\dagger c_{b,j}\rangle c_{a,j}^\dagger c_{a,j} - \langle c_{a,j}^\dagger c_{b,j}\rangle c_{b,j}^\dagger c_{a,j} - \langle c_{b,j}^\dagger c_{a,j}\rangle c_{a,j}^\dagger c_{b,j} + \text{konst.} \tag{22}$$

od tod sledi

$$H_V^{\text{MF}} = Vn_b\sum_j c_{a,j}^\dagger c_{a,j} + Vn_a\sum_j c_{b,j}^\dagger c_{b,j} - V\phi\sum_j c_{b,j}^\dagger c_{a,j} - V\phi^*\sum_j c_{a,j}^\dagger c_{b,j} + NV|\phi|^2 \tag{23}$$

kjer smo definirali:

$$n_a = \langle c_{a,j}^\dagger c_{a,j}\rangle = \frac{1}{N}\sum_{\mathbf{k}}\langle c_{a,\mathbf{k}}^\dagger c_{a,\mathbf{k}}\rangle \tag{24}$$

$$n_b = \langle c_{b,j}^\dagger c_{b,j}\rangle = \frac{1}{N}\sum_{\mathbf{k}}\langle c_{b,\mathbf{k}}^\dagger c_{b,\mathbf{k}}\rangle \tag{25}$$

$$\phi = \langle c_{a,j}^\dagger c_{b,j}\rangle = \frac{1}{N}\sum_{\mathbf{k}}\langle c_{a,\mathbf{k}}^\dagger c_{b,\mathbf{k}}\rangle \tag{26}$$

$$\Delta = -\frac{V}{N}\sum_{\mathbf{k}}\langle c_{a,\mathbf{k}}^\dagger c_{b,\mathbf{k}}\rangle, \quad \Delta^* = -\frac{V}{N}\sum_{\mathbf{k}}\langle c_{b,\mathbf{k}}^\dagger c_{a,\mathbf{k}}\rangle \tag{27}$$

$$\Delta_{\mathbf{k}} = \frac{1}{N}\sum_{\mathbf{k}'} V(\mathbf{k}-\mathbf{k}')\langle a^\dagger_{\mathbf{k}'} b_{\mathbf{k}'}\rangle, \quad V(\mathbf{q}) = \int e^{-i\mathbf{q}\cdot\mathbf{r}} V(\mathbf{r})\ \mathrm{d}\mathbf{r} \tag{28}$$

Uporabimo FT še na celotnem Hamiltonjanu

$$c^\dagger_{a,j} = \frac{1}{\sqrt{N}}\sum_{\mathbf{k}} e^{-i\mathbf{k}\cdot\mathbf{r}_j} c^\dagger_{a,\mathbf{k}} \tag{29}$$

$$H^{\mathrm{MF}} = \sum_{\mathbf{k}} \left([\varepsilon_{a,\mathbf{k}} + V n_b] c^\dagger_{a,\mathbf{k}} c_{a,\mathbf{k}} + [\varepsilon_{b,\mathbf{k}} + V n_a] c^\dagger_{b,\mathbf{k}} c_{b,\mathbf{k}} + \Delta^* c^\dagger_{a,\mathbf{k}} c_{b,\mathbf{k}} + \Delta c^\dagger_{b,\mathbf{k}} c_{a,\mathbf{k}} \right) + \left(\frac{|\Delta|^2}{V} N - V N n_a n_b \right) \tag{30}$$

oziroma:

$$H^{\mathrm{MF}} = \sum_{\mathbf{k}} \begin{pmatrix} c^\dagger_{a,\mathbf{k}} & c^\dagger_{b,\mathbf{k}} \end{pmatrix} \begin{pmatrix} \varepsilon^H_{a,\mathbf{k}} & \Delta^* \\ \Delta & \varepsilon^H_{b,\mathbf{k}} \end{pmatrix} \begin{pmatrix} c_{a,\mathbf{k}} \\ c_{b,\mathbf{k}} \end{pmatrix} + \text{konst.}\,, \tag{31}$$

kjer definiramo Hartree energiji $\varepsilon^H_{a(b),k} = \varepsilon_{a(b),k} + V n_{b(a)}$. Transformacija v diagonalno / Bogoljubov bazo je podobna kot pri BCS, le da ne meša kreatorjev in anihilatorjev / ohranja število delcev:

$$\begin{pmatrix} c_{a,\mathbf{k}} \\ c_{b,\mathbf{k}} \end{pmatrix} = \begin{pmatrix} u_{\mathbf{k}} & v_{\mathbf{k}} \\ -v_{\mathbf{k}} & u_{\mathbf{k}} \end{pmatrix} \begin{pmatrix} \alpha_{\mathbf{k}} \\ \beta_{\mathbf{k}} \end{pmatrix} \tag{32}$$

od tod po enakem postopku kot prej dobimo energiji kvazidelcev

$$E_{\alpha(\beta),\mathbf{k}} = \frac{\varepsilon^H_{a,\mathbf{k}} + \varepsilon^H_{b,\mathbf{k}}}{2} \pm \sqrt{\left(\frac{\varepsilon^H_{a,\mathbf{k}} - \varepsilon^H_{b,\mathbf{k}}}{2} \right)^2 + |\Delta|^2} := \eta_{\mathbf{k}} \pm \sqrt{\xi^2_{\mathbf{k}} + \Delta^2} \tag{33}$$

Za bookkeeping:

$$\left(\frac{\varepsilon^H_a - \varepsilon^H_b}{2} \right) \sin\theta_{\mathbf{k}} + \Delta \cos\theta_{\mathbf{k}} = 0 \implies \tan\theta_{\mathbf{k}} = \frac{-2\Delta}{\varepsilon^H_a - \varepsilon^H_b} = \frac{-\Delta}{\xi_{\mathbf{k}}} \tag{34}$$

$$u_{\mathbf{k}} v_{\mathbf{k}} = -\frac{\sin\theta_{\mathbf{k}}}{2}\,. \tag{35}$$

Za samousklajeno enačbo zdaj potrebujemo

$$\langle c^\dagger_{a,\mathbf{k}} c_{b,\mathbf{k}}\rangle = \langle (u_{\mathbf{k}}\alpha^\dagger_{\mathbf{k}} + v_{\mathbf{k}}\beta^\dagger_{\mathbf{k}})(-v_{\mathbf{k}}\alpha_{\mathbf{k}} + u_{\mathbf{k}}\beta_{\mathbf{k}})\rangle = u_{\mathbf{k}} v_{\mathbf{k}} \left(\langle \beta^\dagger_{\mathbf{k}}\beta_{\mathbf{k}}\rangle - \langle \alpha^\dagger_{\mathbf{k}}\alpha_{\mathbf{k}}\rangle \right) = -\frac{1}{2}\sin\theta_{\mathbf{k}} (f(E_\beta) - f(E_\alpha)) \tag{36}$$

od tod

$$\Delta = -\frac{V}{N}\sum_{\mathbf{k}} \langle c^\dagger_{a,\mathbf{k}} c_{b,\mathbf{k}}\rangle = \frac{V}{N}\sum_{\mathbf{k}} \frac{\Delta}{2\sqrt{\xi^2_{\mathbf{k}} + |\Delta|^2}} (f(E_\beta) - f(E_\alpha)) \tag{37}$$

Opomba, lahko definiramo tudi $n = n_a + n_b = \text{konst.} = 1$ in $n_a - n_b =: m$ od tod sledi

$$H^{\mathrm{MF}} = \sum_{\mathbf{k}} \left(\left[\varepsilon_{a,\mathbf{k}} + \frac{V}{2} n - \frac{V}{2} m \right] c^\dagger_{a,\mathbf{k}} c_{a,\mathbf{k}} + \left[\varepsilon_{b,\mathbf{k}} + \frac{V}{2} n + \frac{V}{2} m \right] c^\dagger_{b,\mathbf{k}} c_{b,\mathbf{k}} + \Delta^* c^\dagger_{a,\mathbf{k}} c_{b,\mathbf{k}} + \Delta c^\dagger_{b,\mathbf{k}} c_{a,\mathbf{k}} \right) \tag{38}$$

od koder je v 2D Hartree gap enak:

$$G^H = G - 4(t_a - t_b) - V m\,. \tag{39}$$

Note: v praksi je $-1 < m < 0$, več elektronov je v valenčnem pasu. V dveh dimenzijah se Hartree pasova srečata na energiji, ki jo lahko definiramo prek $\xi_{\mathbf{k}} = 0$ v semimetal primeru. Sicer pa je to enako zamiku pasov, ki še ne hibiridizirata:

$$E_F = \frac{V}{2} n - \frac{t_a + t_b}{t_a - t_b} \frac{G - V m}{2}\,. \tag{40}$$

Zamik je enak $V/2$, ko sta osnovna pasova simetrična in je zasedenost polovična. V primeru simetričnih pasov sta disperziji kvazidelcev

$$E_{\alpha(\beta),\mathbf{k}} - V/2 = \pm\sqrt{\left(\frac{\epsilon_{a,\mathbf{k}} - \epsilon_{b,\mathbf{k}} - Vm}{2}\right)^2 + |\Delta|^2}\,. \tag{41}$$

Velja še

$$n_a = \frac{1}{N}\sum_{\mathbf{k}} \left(u_{\mathbf{k}}^2 f(E_\alpha) + v_{\mathbf{k}}^2 f(E_\beta)\right) \tag{42}$$

$$n_b = \frac{1}{N}\sum_{\mathbf{k}} \left(v_{\mathbf{k}}^2 f(E_\alpha) + u_{\mathbf{k}}^2 f(E_\beta)\right) \tag{43}$$

$$u_{\mathbf{k}}^2 = \frac{1}{2}\left(1 + \frac{\xi_{\mathbf{k}}}{\sqrt{\Delta^2 + \xi_{\mathbf{k}}^2}}\right) \tag{44}$$

$$v_{\mathbf{k}}^2 = \frac{1}{2}\left(1 - \frac{\xi_{\mathbf{k}}}{\sqrt{\Delta^2 + \xi_{\mathbf{k}}^2}}\right) \tag{45}$$

V termičnem stanju (equlibrium case), rešujemo sistem treh enačb. Prva za razliko okupacij

$$m = n_a - n_b = \frac{1}{N}\sum_{\mathbf{k}} \frac{\xi_{\mathbf{k}}}{\sqrt{\xi_{\mathbf{k}}^2 + \Delta^2}}(f(E_{\alpha,\mathbf{k}}) - f(E_{\beta,\mathbf{k}}))\,, \tag{46}$$

druga za ohranitev števila delev/kvazidelcev:

$$1 = n = n_a + n_b = \frac{1}{N}\sum_{\mathbf{k}}(f(E_{\alpha,\mathbf{k}}) + f(E_{\beta,\mathbf{k}}))\,, \tag{47}$$

in enačbo reže:

$$\frac{1}{V} = \frac{1}{N}\sum_{\mathbf{k}} \frac{f(E_{\beta,\mathbf{k}}) - f(E_{\alpha,\mathbf{k}})}{2\sqrt{\xi_{\mathbf{k}}^2 + \Delta^2}}\,, \tag{48}$$

kjer dobimo rešitev za par m, Δ glede na izbrane parametre.

1.3 Numerika/psevdokoda

1.4 Odprti sistemi in modeliranje kopeli

Born-Markov master enačbe in Redfield: ločimo kopel in sistem,

$$\rho_S(t) = \mathrm{Tr}_B(\rho(t)) \tag{49}$$

Hamiltonjan razcepimo na $H = H_0 + H_I$, kjer je $H_0 = H_S \otimes \mathbb{I}_B + \mathbb{I}_S \otimes H_B$ razdenjen na podsistema, za interakcijski del pa privzamemo obliko $H_I = \sum_\alpha A_\alpha \otimes B_\alpha$. V interakcijski sliki je operator za evolucijo

$$U_0(t, t_0) = e^{-iH_0(t-t_0)} \tag{50}$$

$$\rho_I(t) = U_0^\dagger(t,0)\rho(t)U_0(t,0) = e^{iH_0 t}\rho(t)e^{-iH_0 t} \tag{51}$$

$$H_I(t) = U_0^\dagger(t,0)H_I U_0(t,0) = e^{iH_0 t}H_I e^{-iH_0 t} \tag{52}$$

Von Neumannova enačba je

$$\frac{d}{dt}\rho_I(t) = -i[H_I(t), \rho_I(t)] \tag{53}$$

$$\rho_I(t) = \rho_I(0) - i\int_0^t dt'\,[H_I(t'), \rho_I(t')] \tag{54}$$

vstavimo nazaj v komutator in se znebimo prostosnih stopenj kopeli:

$$\frac{d}{dt}\rho_S(t) = -\int_0^t ds\, \mathrm{Tr}_B\left([H_I(t),[H_I(t-s),\rho_S(t)\otimes\rho_B]]\right) \tag{55}$$

V interakcijskem členu so operatorji A_α napisani kot lastni operatorji H_S z energijami prehodov $\omega = E_b - E_a$

$$A_\alpha(\omega) = \sum_{E_b - E_a = \omega} |a\rangle\langle a|A_\alpha|b\rangle\langle b| \tag{56}$$

To zadošča relacijam $[H_S, A_\alpha(\omega)] = -\omega A_\alpha(\omega), \quad [H_S, A_\alpha^\dagger(\omega)] = \omega A_\alpha^\dagger(\omega)$ in
$e^{iH_S t}A_\alpha(\omega)e^{-iH_S t} = e^{-i\omega t}A_\alpha(\omega), \quad e^{iH_S t}A_\alpha^\dagger(\omega)e^{-iH_S t} = e^{i\omega t}A_\alpha^\dagger(\omega)$. V interakcijski sliki potem velja

$$H_I(t) = \sum_{\alpha,\omega} e^{-i\omega t}A_\alpha(\omega)\otimes B_\alpha(t) = e^{iH_B t}B_\alpha e^{-iH_B t} \tag{57}$$

evolucija sledi kot

$$\dot\rho_S = \sum_{\omega,\omega'}\sum_{\alpha,\beta} e^{i(\omega'-\omega)t}\Gamma_{\alpha\beta}(\omega)\left(A_\alpha(\omega)\rho_S(t)A_\beta^\dagger(\omega') - A_\beta^\dagger(\omega')A_\alpha(\omega)\rho_S(t)\right) + \text{h.c.} \tag{58}$$

kjer so prehodne matrike med stanji

$$\Gamma_{\alpha\beta}(\omega) = \int_0^\infty ds\, e^{i\omega s}\,\mathrm{Tr}_B\left(B_\alpha^\dagger(t)B_\beta(t-s)\rho_B\right) = \frac{1}{2}\gamma_{\alpha\beta}(\omega) + iS_{\alpha\beta}(\omega) \tag{59}$$

Evolucija pa ima strukturo

$$\frac{d\rho_S}{dt} = -i[H_{LS},\rho_S(t)] + \mathcal{D}(\rho_S(t)) \tag{60}$$

z disipatorjem

$$\mathcal{D}(\rho_S) = \sum_\omega\sum_{\alpha,\beta}\gamma_{\alpha\beta}(\omega)\left(A_\beta(\omega)\rho_S A_\alpha^\dagger(\omega) - \frac{1}{2}\{A_\alpha^\dagger(\omega)A_\beta(\omega),\rho_S\}\right) \tag{61}$$

oz. v splošnem

$$\mathcal{D}(\rho_S) = \sum_k \gamma_k\left(L_k\rho_S L_k^\dagger - \frac{1}{2}\{L_k^\dagger L_k,\rho_S\}\right) \tag{62}$$

V stacionarnem stanju velja

$$\mathcal{L}\rho_{\text{ness}} = 0\,. \tag{63}$$

Notes za vektorizacijo in superoperatorje:

$$\mathcal{L}u_\mu^R = \lambda_\mu u_\mu^R \tag{64}$$

$$\mathrm{vec}(ABC) = (A\otimes C^T)\mathrm{vec}(B) \tag{65}$$

$$\mathrm{Tr}(A^\dagger B) = \mathrm{vec}(A)^\dagger\mathrm{vec}(B) \tag{66}$$

$$\frac{d}{dt}|\rho\rangle = \left(-iH\otimes\mathbb{I} + i\mathbb{I}\otimes H^T + \sum_k L_k\otimes L_k^* - \frac{1}{2}\sum_k(L_k^\dagger L_k\otimes\mathbb{I} + \mathbb{I}\otimes L_k^T L_k^*)\right)|\rho\rangle \tag{67}$$

$$\langle\hat O\rangle(t) = \langle\mathbb{I}|\hat O\otimes\mathbb{I}|\rho(t)\rangle \tag{68}$$

Šibke sklopitve in perturbacijski pristopi

$$\mathcal{L} = -i[H, \cdot] + \epsilon\mathcal{D}[\cdot], \qquad \rho = \rho_0 + \epsilon\rho_1 + \epsilon^2\rho_2 + \dots \tag{69}$$

napišemo gostotno matriko sistema v lastni bazi, siagonalno, s klasičnimi populacijami

$$\rho_0 = \sum_n p_n|n\rangle\langle n|, \quad H|n\rangle = E_n|n\rangle \tag{70}$$

kjer velja v ničtem redu (by construction)

$$\mathcal{L}\rho_0 = -i[H, \rho_0] = 0 \tag{71}$$

V prvem redu so enačbe za NESS

$$-i[H, \rho_1] + \mathcal{D}(\rho_0) = 0 \tag{72}$$

kjer je $\mathcal{D}(\rho_0) = \sum_\mu \left(L_\mu\rho_0 L_\mu^\dagger - \frac{1}{2}\{L_\mu^\dagger L_\mu, \rho_0\}\right)$ in od tod enačba za prehode med populacijami

$$\langle n|\mathcal{D}(\rho_0)|n\rangle = \sum_\mu\sum_m p_m|\langle n|L_\mu|m\rangle|^2 - \sum_\mu\sum_m p_n|\langle m|L_\mu|n\rangle|^2 \tag{73}$$

kot klasična populacijska dinamika z $R_{nm} = \sum_\mu |\langle n|L_\mu|m\rangle|^2$

$$\dot{p}_n = \epsilon\sum_m (R_{nm}p_m - R_{mn}p_n) \tag{74}$$

Izvendiagonali elementi se razvijejo hitro na $\mathcal{O}(1)$, medtem ko se populacije razvijajo počasi na GGE manifoldu z $\mathcal{O}(\epsilon^{-1})$. Za ohranitev normalizacije mora veljati $\sum_n \dot{p}_n = 0$.

Integrabilni sistemi imajo makrsoskopsko število ohranjenih nabojev $[H_0, C_i] = 0$, pod unitarno časovno evolucijo se razvijejo v netermično stanje

$$\rho_{GGE} = \frac{1}{Z}e^{-\sum_i \lambda_i C_i} \tag{75}$$

$$\langle\psi_0|C_i|\psi_0\rangle = \mathrm{Tr}(C_i\rho_{\mathrm{GGE}}) \tag{76}$$

fiksira λ_i. UNitarna dinamika prijcira na GGE manifold, šibka sklopitev premika po GGE manifoldu:

$$\rho_0(t) = \rho_{\boldsymbol{\lambda}(t)} = \frac{1}{Z(t)}\exp\left(-\sum_i \lambda_i(t)C_i\right) \tag{77}$$

EOM za ohranjene naboje je

$$\langle\dot{C}_i\rangle_{\lambda(\infty)} = \mathrm{Tr}[C_i\mathcal{D}\rho] \simeq \epsilon\,\mathrm{Tr}[C_i\mathcal{D}\rho_0] = 0 \quad \forall\lambda_i \tag{78}$$

Definiramo projektor na GGE manifold:

$$PX = -\sum_{ij}\frac{\partial\rho}{\partial\lambda_i}(\chi^{-1})_{ij}\mathrm{Tr}[C_j X] \tag{79}$$

ki da dinamiko vzdolž manifolda:

$$\frac{\partial\rho_{\boldsymbol{\lambda}}}{\partial t} = \sum_i \frac{\partial\rho_{\boldsymbol{\lambda}}}{\partial\lambda_i}\dot{\lambda}_i = \epsilon P\mathcal{D}(\rho_{\boldsymbol{\lambda}}) = -\epsilon\sum_{ij}\frac{\partial\rho_{\boldsymbol{\lambda}}}{\partial\lambda_i}(\chi^{-1})_{ij}\mathrm{Tr}\left[C_j\mathcal{D}(\rho_{\boldsymbol{\lambda}})\right] \tag{80}$$

od koder sledi evolucija za koeficiente/potenciale λ_i:

$$\dot{\lambda}_i = -\epsilon\sum_j(\chi^{-1})_{ij}\mathrm{Tr}[C_j\mathcal{D}\rho_{\lambda(t)}] \tag{81}$$

s susceptibilnostmi $\chi_{ij} = \langle C_iC_j\rangle - \langle C_i\rangle\langle C_j\rangle = -\mathrm{Tr}\left[C_i\frac{\partial\rho}{\partial\lambda_j}\right]$.

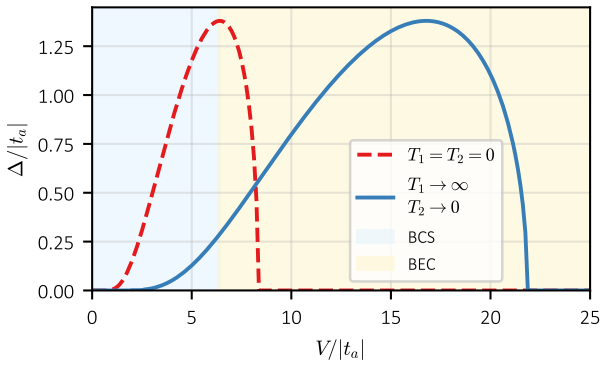
2 Ekscitonski izolatorji in sklopitev s kopelmi

$$\mathrm{Tr}\left[f\frac{1}{2}\{\rho,A\}\right] = \mathrm{Tr}\left[\frac{1}{2}f\rho A + \frac{1}{2}fA\rho\right] = \frac{1}{2}\mathrm{Tr}\left[Af\rho + fA\rho\right] \tag{82}$$

$$= \frac{1}{2}\mathrm{Tr}\left[(Af+fA)\rho\right] = \frac{1}{2}\langle Af+fA\rangle \tag{83}$$

2.1 Osnovni modeli kopeli

2.2 Limita neskončne temperature



Slika 1 | Gap pri $T_1 \to \infty$ in $T_2 \to 0$.

2.3 Fononske sklopitve

Najprej pogledamo, kako modificiran (Peierls tip) hopping integral in modificiran orbital energy vplivata na sklopitev. Model spremenimo: orbitali znotraj o.c. sta na enakem položaju in se deformirata vedno v isto smer, torej $\mathbf{u}_{a,i} = \mathbf{u}_{b,i} = \mathbf{u}_i$. Coulombski potencial se zaradi deformacije ne spremeni veliko. Hamiltonka za nedeformiran kristal ostane enaka, višje člene pa obravnavamo perturbativno:

$$H[\{\mathbf{u}\}] = \underbrace{\sum_{i,\gamma,\gamma'} \varepsilon_{\gamma\gamma',i}[\{\mathbf{u}\}]\, c^\dagger_{\gamma,i}c_{\gamma',i}}_{\text{on-site in CEF}} - \underbrace{\sum_{\langle i,j\rangle,\gamma} t_\gamma(|\mathbf{r}_{ij}|)\, c^\dagger_{\gamma,i}c_{\gamma,j}}_{\text{hopping, med oribitalami}} + \underbrace{V\sum_j n_{a,j}n_{b,j}}_{\text{Coulomb}} \tag{84}$$

Razdalje so:

$$\mathbf{r}_j = \mathbf{R}_j + \mathbf{u}_j\,, \tag{85}$$

oziroma razdalje med paroma (sosedi):

$$\mathbf{r}_{ij} = \boldsymbol{\delta} + \mathbf{u}_j - \mathbf{u}_i \;\to |\mathbf{r}_{ij}| \simeq |\boldsymbol{\delta}| + \boldsymbol{\delta}\cdot(\mathbf{u}_j - \mathbf{u}_i) + \mathcal{O}(u^2) \tag{86}$$

Predpostavimo da na hopping vpliva samo sprememba razdalje (delam v enotah, kjer je ravnovesna razdalja med atomi $a_0 = 1$):

$$t_\gamma(|\mathbf{r}_{ij}|) \simeq t_\gamma(|\boldsymbol{\delta}|) + \boldsymbol{\delta}\cdot(\mathbf{u}_j - \mathbf{u}_i)\left.\frac{\mathrm{d}t_\gamma(x)}{\mathrm{d}x}\right|_{x=|\boldsymbol{\delta}|=1} \tag{87}$$

Deformacija vpliva na vezavno energijo preko CEF, razvijemo preko deformacijskega tenzorja ($\varepsilon_{\mu\nu} = \frac{1}{2}(\partial_\mu u_\nu + \partial_\nu u_\mu)$ oziroma diskretno: $\varepsilon_{\mu\nu,j} = \frac{1}{4}(u^\nu_{j+\mu} - u^\nu_{j-\mu} + u^\mu_{j+\nu} - u^\mu_{j-\nu})$ zgornji indeks je koordinata vektorja). Če je eno elektronska Hamiltonka na mestu j (el. - ion) (po μ,ν se sešteva po Einsteinu):

$$h_j = h_0 + \delta h_j + O(\varepsilon^2), \quad \delta h_j = \frac{\partial U_{\text{el-ion}}}{\partial \varepsilon_{\mu\nu,j}}\varepsilon_{\mu\nu,j},\ (\mu,\nu) = (x,y) \tag{88}$$

in je v nedeformiranem stanju:

$$\varepsilon_\gamma^{(0)} = \langle \phi_\gamma | h_0 | \phi_\gamma \rangle \tag{89}$$

potem so popravki (lahko mešajo orbitali)

$$\delta\varepsilon_{\gamma\gamma'} = \left\langle \phi_\gamma \left| \frac{\partial U_{\text{el-ion}}}{\partial \varepsilon_{\mu\nu,j}} \right| \phi_{\gamma'} \right\rangle \varepsilon_{\mu\nu,j} = D^{\mu\nu}_{\gamma\gamma'} \varepsilon_{\mu\nu,j} \tag{90}$$

Sklopitvena Hamiltonka v realni bazi je tako ($\sum_{\langle ij \rangle} \to \sum_i \sum_{\delta=\{x,y\}}$)

$$H_{ep}^{(1)} = \sum_{i,\gamma,\gamma'} D^{\mu\nu}_{\gamma\gamma'} \varepsilon_{\mu\nu,i} c^\dagger_{\gamma,i} c_{\gamma',i} - \sum_{i,\delta,\gamma} t'_\gamma \boldsymbol{\delta} \cdot (\mathbf{u}_{i+\delta} - \mathbf{u}_i) \left(c^\dagger_{\gamma,i} c_{\gamma,i+\delta} + \text{h.c.} \right) \tag{91}$$

(note):

$$\sum_{\boldsymbol{\delta}} \boldsymbol{\delta} \cdot (\mathbf{u}_{i+\delta} - \mathbf{u}_i) = \hat{x} \cdot (\mathbf{u}_{i+x} - \mathbf{u}_i) - \hat{x} \cdot (\mathbf{u}_{i-x} - \mathbf{u}_i) + \hat{y} \cdot (\mathbf{u}_{i+y} - \mathbf{u}_i) - \hat{y} \cdot (\mathbf{u}_{i-y} - \mathbf{u}_i) \approx 2(\varepsilon_{xx} + \varepsilon_{yy}) \tag{92}$$

Zdaj kvantiziramo odmike in prepišemo v k-prostor, specificno za 2D square lattice:

$$\mathbf{u}_j = \frac{1}{\sqrt{N}} \sum_{\mathbf{q},\lambda} \frac{1}{\sqrt{2M\omega_{\mathbf{q}\lambda}}} \boldsymbol{\epsilon}_{\mathbf{q}\lambda} e^{i\mathbf{q}\cdot\mathbf{R}_j} \left(b_{\mathbf{q}\lambda} + b^\dagger_{-\mathbf{q}\lambda} \right) \tag{93}$$

v prvem členu so sami členi oblike (od defomracijskega tenzorja, diskretna verzija)

$$u^\mu_{j+\delta} - u^\mu_{j-\delta} = \frac{2i}{\sqrt{N}} \sum_{\mathbf{q},\lambda} \frac{\epsilon^\mu_{\mathbf{q}\lambda}}{\sqrt{2M\omega_{\mathbf{q}\lambda}}} \sin(q_\delta)\, e^{i\mathbf{q}\cdot\mathbf{R}_j} \left(b_{\mathbf{q}\lambda} + b^\dagger_{-\mathbf{q}\lambda} \right). \tag{94}$$

v drugem pa členi ($\delta = \mu = \{x, y\}$, glej note):

$$\frac{t'_\gamma}{\sqrt{N}} \sum_{\mathbf{q},\mathbf{k}',\mathbf{k}} \frac{\epsilon^\mu_{\mathbf{q}\lambda}}{\sqrt{2M\omega_{\mathbf{q}\lambda}}} \left(b_{\mathbf{q}\lambda} + b^\dagger_{-\mathbf{q}\lambda} \right) \left(e^{iq_\mu} - 1 \right) \left(e^{ik_\mu} + e^{-ik'_\mu} \right) c^\dagger_{\gamma,\mathbf{k}'} c_{\gamma,\mathbf{k}} \sum_j \frac{1}{N} e^{i(\mathbf{q}+\mathbf{k}-\mathbf{k}')\cdot\mathbf{R}_j} = \tag{95}$$

$$= \frac{t'_\gamma}{\sqrt{N}} \sum_{\mathbf{k},\mathbf{q}} \frac{\epsilon^\mu_{\mathbf{q}\lambda}}{\sqrt{2M\omega_{\mathbf{q}\lambda}}} \left(b_{\mathbf{q}\lambda} + b^\dagger_{-\mathbf{q}\lambda} \right) \left(e^{iq_\mu} - 1 \right) \left(e^{ik_\mu} + e^{-i(k_\mu+q_\mu)} \right) c^\dagger_{\gamma,\mathbf{k}+\mathbf{q}} c_{\gamma,\mathbf{k}} = \tag{96}$$

$$= \frac{2it'_\gamma}{\sqrt{N}} \sum_{\mathbf{k},\mathbf{q}} \frac{\epsilon^\mu_{\mathbf{q}\lambda}}{\sqrt{2M\omega_{\mathbf{q}\lambda}}} \left(\sin(k_\mu + q_\mu) - \sin(k_\mu) \right) c^\dagger_{\gamma,\mathbf{k}+\mathbf{q}} c_{\gamma,\mathbf{k}} \left(b_{\mathbf{q}\lambda} + b^\dagger_{-\mathbf{q}\lambda} \right) \tag{97}$$

Hamiltonka je na koncu oblike (intraband = $\gamma = \gamma'$):

$$H_{ep} = \frac{1}{\sqrt{N}} \sum_{\mathbf{k},\mathbf{q},\lambda} \psi^\dagger_{\mathbf{k}+\mathbf{q}} \begin{pmatrix} g^\lambda_{aa}(\mathbf{k},\mathbf{q}) & g^\lambda_{ab}(\mathbf{q}) \\ g^\lambda_{ba}(\mathbf{q}) & g^\lambda_{bb}(\mathbf{k},\mathbf{q}) \end{pmatrix} \psi_{\mathbf{k}} \left(b_{\mathbf{q}\lambda} + b^\dagger_{-\mathbf{q}\lambda} \right) \tag{98}$$

$$g^\lambda(\mathbf{k},\mathbf{q}) = \frac{i}{\sqrt{2M\omega_{\mathbf{q}\lambda}}} \begin{pmatrix} -2t'_a T^\lambda(\mathbf{k},\mathbf{q}) + F^\lambda_{aa}(\mathbf{q}) & F^\lambda_{ab}(\mathbf{q}) \\ F^\lambda_{ba}(\mathbf{q}) & -2t'_b T^\lambda(\mathbf{k},\mathbf{q}) + F^\lambda_{bb}(\mathbf{q}) \end{pmatrix}. \tag{99}$$

kjer so:

$$T^\lambda(\mathbf{k},\mathbf{q}) = \sum_{\mu=x,y} \epsilon^\mu_{\mathbf{q}\lambda} \left[\sin(k_\mu + q_\mu) - \sin(k_\mu) \right] \tag{100}$$

$$F^\lambda_{\gamma\gamma'}(\mathbf{q}) = D^{xx}_{\gamma\gamma'} \epsilon^x_{\mathbf{q}\lambda} \sin(q_x) + D^{yy}_{\gamma\gamma'} \epsilon^y_{\mathbf{q}\lambda} \sin(q_y) + D^{xy}_{\gamma\gamma'} \left[\epsilon^y_{\mathbf{q}\lambda} \sin(q_x) + \epsilon^x_{\mathbf{q}\lambda} \sin(q_y) \right]. \tag{101}$$

interband členi so neničelni, glede na izbrane orbitale (glej (90)) in izbran CF potencial, odvisno od simetrije. Ker smo izbrali $\mathbf{u}_{a,i} = \mathbf{u}_{b,i}$ se ti elementi sklapljajo samo z akustičnimi fononi, ce znotraj o.c. ni prostostnih stopenj.

2.4 Optični fononi

Rabimo dodatne ligande / dodatne ione znotraj celice. Še vedno obdržimo $\mathbf{u}_{a,j} = \mathbf{u}_{b,j} = \mathbf{u}_j$, okolica pa se deformira neodvisno. Odmik liganda ℓ v celici j razvijemo po normalnih nacinih

$$\mathbf{v}_{j,\ell} = \sum_s Q_{s,j}\, \mathbf{e}_{s,\ell}, \qquad \sum_\ell \mathbf{e}_{s,\ell} \cdot \mathbf{e}_{s',\ell} = \delta_{ss'}, \tag{102}$$

kjer je s indeks nacina, $Q_{s,j}$ skalarna amplituda. Razvijemo enoelektronsko hamiltonko

$$h_j = h_0 + \sum_s \left.\frac{\partial U_{\text{el-ion}}}{\partial Q_{s,j}}\right|_{Q=0} Q_{s,j} + O(Q^2), \qquad \Lambda^s_{\gamma\gamma'} = \left\langle \phi_\gamma \left| \frac{\partial U_{\text{el-ion}}}{\partial Q_{s,j}} \right| \phi_{\gamma'} \right\rangle_0 . \tag{103}$$

$$H^{\text{opt}}_{ep} = \sum_{j,s} \sum_{\gamma,\gamma'} \Lambda^s_{\gamma\gamma'} Q_{s,j}\, c^\dagger_{\gamma,j} c_{\gamma',j}. \tag{104}$$

Kvantiziramo koordinate okoliških ionov:

$$Q_{s,j} = \frac{1}{\sqrt{N}} \sum_{\mathbf{q}} \frac{1}{\sqrt{2M_s\omega_{\mathbf{q}s}}} e^{i\mathbf{q}\cdot\mathbf{R}_j} \left(b_{\mathbf{q}s} + b^\dagger_{-\mathbf{q}s} \right), \tag{105}$$

od tod

$$H^{\text{opt}}_{ep} = \frac{1}{\sqrt{N}} \sum_{\mathbf{k},\mathbf{q},s} \psi^\dagger_{\mathbf{k}+\mathbf{q}}\, g^s(\mathbf{q})\, \psi_{\mathbf{k}} \left(b_{\mathbf{q}s} + b^\dagger_{-\mathbf{q}s} \right), \qquad g^s(\mathbf{q}) = \frac{\Lambda^s}{\sqrt{2M_s\omega_{\mathbf{q}s}}}. \tag{106}$$

Kar da končne sklopitve pri $\mathbf{q} \to 0$ ($\Omega_s \equiv \omega_{\mathbf{q}\to 0,s}$):

$$g^s(\mathbf{q} \to 0) = \frac{\Lambda^s}{\sqrt{2M_s\Omega_s}}. \tag{107}$$

2.5 Fotonske sklopitve v resonatorju

Začnemo z minimalnim modelom za elektrone, n.n. hopping + lokalni Coulumb:

$$H_e = H^{(0)}_e + H_V = \sum_{\mathbf{k}} \varepsilon_{a,\mathbf{k}} c^\dagger_{a,\mathbf{k}} c_{a,\mathbf{k}} + \sum_{\mathbf{k}} \varepsilon_{b,\mathbf{k}} c^\dagger_{b,\mathbf{k}} c_{b,\mathbf{k}} + V \sum_j c^\dagger_{a,j} c_{a,j} c^\dagger_{b,j} c_{b,j} \tag{108}$$

in privzamemo 2D plast, kjer sta dve orbitali na o.c., ki hibridizirata samo preko Coulombske interakcije. Disperzije za model neinteragirajoč sistem so v tem primeru enake (21). Interakcija pa je v prvem redu oblike:

$$H^{(1)}_{e\gamma} = -\int \mathrm{d}\mathbf{r}\, \mathbf{j}_e \cdot \mathbf{A} = -\int \mathrm{d}\mathbf{r} \sum_{\mathbf{q}} \mathbf{A}(\mathbf{q}) e^{i\mathbf{q}\cdot\mathbf{r}} \mathbf{j}_e(\mathbf{r}) = -\sum_{\mathbf{q}} \mathbf{A}(\mathbf{q}) \cdot \mathbf{j}_e(-\mathbf{q}), \tag{109}$$

kjer je

$$\mathbf{A}(\mathbf{q}) = \sum_\lambda \frac{1}{\sqrt{2\mathcal{V}\epsilon_0\omega_{\mathbf{q},\lambda}}} \boldsymbol{\epsilon}_{\mathbf{q},\lambda} \left(a_{\mathbf{q},\lambda} + a^\dagger_{-\mathbf{q},\lambda} \right) \tag{110}$$

V t.b. modelu dobimo tok iz kontinuitetne in Heisenbergove enačbe:

$$\frac{\partial \rho(\mathbf{r})}{\partial t} + \nabla \cdot \mathbf{j} = 0 \;\to\; i\mathbf{q} \cdot \mathbf{j}(\mathbf{q}) = -\frac{\partial \rho(\mathbf{q})}{\partial t} \tag{111}$$

(+ predznak je od per partes intergacije) in

$$\mathbf{q} \cdot \mathbf{j}_e(\mathbf{q}) = i\frac{\partial \rho(\mathbf{q})}{\partial t} = -[H_e, \rho(\mathbf{q})] \tag{112}$$

Elektronska gostota je

$$\rho(\mathbf{r}) = -e \cdot \psi^\dagger(\mathbf{r})\psi(\mathbf{r}) \tag{113}$$

operatorje polja razvijemo po lokaliziranih / Wannierovih orbitalah

$$\psi(\mathbf{r}) = \sum_{i,\gamma=\{a,b\}} \phi_\gamma(\mathbf{r}-\mathbf{R}_i)c_{\gamma i}\,. \tag{114}$$

Transformiramo s FT, uvedemo novo spremenljivko $\mathbf{r} = \mathbf{R}_i + \mathbf{x}$ in razvijemo do dipolov $e^{-i\mathbf{q}\cdot\mathbf{x}} \simeq 1 - i\mathbf{q}\cdot\mathbf{x}$ (delamo se da se orbitale ne prekrivajo izven o.c., torej $\sum_{ij} \to \sum_i \delta_{ij}$):

$$\rho(\mathbf{q}) = -e\sum_{i,j}\sum_{\gamma,\gamma'}\int \mathrm{d}\mathbf{r}\; e^{-i\mathbf{q}\cdot\mathbf{r}}\phi^*_\gamma(\mathbf{r}-\mathbf{R}_i)\phi_{\gamma'}(\mathbf{r}-\mathbf{R}_j)c^\dagger_{\gamma i}c_{\gamma' j} = -e\sum_{i,\gamma,\gamma'} e^{-i\mathbf{q}\cdot\mathbf{R}_i}\langle\phi_\gamma|e^{-i\mathbf{q}\cdot\mathbf{x}}|\phi_{\gamma'}\rangle c^\dagger_{\gamma i}c_{\gamma' i} \tag{115}$$

Zdaj pa recimo da je dipolni moment samo med različnima orbitalama v o.c. in $n_{\gamma i} = \langle\phi_\gamma|\phi_\gamma\rangle c^\dagger_{\gamma,i}c_{\gamma,i}$:

$$\rho(\mathbf{q}) \simeq -e\sum_i e^{-i\mathbf{q}\cdot\mathbf{R}_i}\left(n_{ai} + n_{bi} - i\mathbf{q}\cdot\mathbf{d}\left(c^\dagger_{a,i}c_{b,i} + c^\dagger_{b,i}c_{a,i}\right)\right) \tag{116}$$

zdaj uporabimo zvezo

$$\sum_i e^{i\mathbf{q}\cdot\mathbf{R}_i}c^\dagger_{ai}c_{bi} = \sum_{\mathbf{k}} c^\dagger_{a,\mathbf{k}+\mathbf{q}}c_{b\mathbf{k}} \tag{117}$$

z negativnim argumentom je potem elektronska gostota do dipolnega člena (upoštevajoč $\mathbf{d} = \langle\phi_a(\mathbf{x})|\mathbf{x}|\phi_b(\mathbf{x})\rangle$):

$$\rho(-\mathbf{q}) = -e\sum_{\mathbf{k}}\begin{pmatrix}c^\dagger_{a,\mathbf{k}+\mathbf{q}} & c^\dagger_{b,\mathbf{k}+\mathbf{q}}\end{pmatrix}\left(\mathbb{1} + i\,\mathbf{q}\cdot\mathbf{d}\begin{pmatrix}0&1\\1&0\end{pmatrix}\right)\begin{pmatrix}c_{a\mathbf{k}}\\c_{b\mathbf{k}}\end{pmatrix} \tag{118}$$

Zdaj poračunamo komutator, da dobimo tok. Delam z vektorji $\psi_k = \begin{pmatrix}c_{a,\mathbf{k}} & c_{b,\mathbf{k}}\end{pmatrix}^\top$:

$$[H^{(0)}_e, \rho(\mathbf{q})] = -e\sum_{\mathbf{k},\mathbf{k}'}\left[\psi^\dagger_{\mathbf{k}}\begin{pmatrix}\varepsilon_{a,\mathbf{k}} & 0\\0 & \varepsilon_{b,\mathbf{k}}\end{pmatrix}\psi_{\mathbf{k}}, \psi^\dagger_{\mathbf{k}'-\mathbf{q}}\begin{pmatrix}1 & -i\mathbf{q}\cdot\mathbf{d}\\-i\mathbf{q}\cdot\mathbf{d} & 1\end{pmatrix}\psi_{\mathbf{k}'}\right] \tag{119}$$

uporabimo zvezo:

$$\left[\psi^\dagger_{\mathbf{k}}A\psi_{\mathbf{k}}, \psi^\dagger_{\mathbf{k}'-\mathbf{q}}B\psi_{\mathbf{k}'}\right] = \psi^\dagger_{\mathbf{k}'-\mathbf{q}}\Big(A(\mathbf{k}'-\mathbf{q})B(\mathbf{k}') - B(\mathbf{k}'-\mathbf{q})A(\mathbf{k}')\Big)\psi_{\mathbf{k}'}, \delta_{\mathbf{k},\mathbf{k}'-\mathbf{q}} \tag{120}$$

$$= \psi^\dagger_{\mathbf{k}}\Big(A(\mathbf{k})B(\mathbf{k}+\mathbf{q}) - B(\mathbf{k})A(\mathbf{k}+\mathbf{q})\Big)\psi_{\mathbf{k}+\mathbf{q}} \tag{121}$$

od tod

$$\mathbf{q}\cdot\mathbf{j}_e(\mathbf{q}) = -[H^{(0)}_e, \rho(\mathbf{q})] = -e\sum_{\mathbf{k}}\psi^\dagger_{\mathbf{k}}\begin{pmatrix}\varepsilon_{a,\mathbf{k}+\mathbf{q}} - \varepsilon_{a,\mathbf{k}} & i(\mathbf{q}\cdot\mathbf{d})\left(\varepsilon_{a,\mathbf{k}} - \varepsilon_{b,\mathbf{k}+\mathbf{q}}\right)\\ i(\mathbf{q}\cdot\mathbf{d})\left(\varepsilon_{b,\mathbf{k}} - \varepsilon_{a,\mathbf{k}+\mathbf{q}}\right) & \varepsilon_{b,\mathbf{k}+\mathbf{q}} - \varepsilon_{b,\mathbf{k}}\end{pmatrix}\psi_{\mathbf{k}+\mathbf{q}} \tag{122}$$

da izrazimo tok, naredimo približek, da prispevajo le majhni $\mathbf{q}\to 0$, zato:

$$\varepsilon_{a,\mathbf{k}+\mathbf{q}} - \varepsilon_{a,\mathbf{k}} \simeq \mathbf{q}\cdot\nabla_{\mathbf{k}}\varepsilon_{\mathbf{k}+\mathbf{q}/2}\,, \tag{123}$$

kar drži točno za proste elektrone in približno za tight binding disperzije. In general ce ne expandamo:

$$\mathbf{j}_e(\mathbf{q}) = -\frac{e\mathbf{q}}{\|\mathbf{q}\|^2}\sum_{\mathbf{k}}\psi^\dagger_{\mathbf{k}}M(\mathbf{k},\mathbf{q})\psi_{\mathbf{k}+\mathbf{q}} \;+\; \mathbf{q}\times\mathbf{C}(\mathbf{q}) \tag{124}$$

kjer so $\mathbf{C}(\mathbf{q})$ arbitrarna vektorska polja (tokovi brez divergence, magnetizacijski tokovi). Ce pa naredimo ta priblizek pa dobimo za tok v splošnem, indekse zamaknemo za $-\mathbf{q}$:

$$\mathbf{j}_e(\mathbf{q}) \simeq -e\sum_{\mathbf{k}}\psi^\dagger_{\mathbf{k}-\mathbf{q}}\begin{pmatrix}\nabla_{\mathbf{k}}\varepsilon_{a,\mathbf{k}-\mathbf{q}/2} & i\mathbf{d}(\varepsilon_{a,\mathbf{k}-\mathbf{q}} - \epsilon_{b\mathbf{k}})\\ i\mathbf{d}(\varepsilon_{b,\mathbf{k}-\mathbf{q}} - \varepsilon_{a\mathbf{k}}) & \nabla_{\mathbf{k}}\varepsilon_{b,\mathbf{k}-\mathbf{q}/2}\end{pmatrix}\psi_{\mathbf{k}}\,. \tag{125}$$

Od tod splošna Hamiltonka v prvem redu:

$$H_{e\gamma}^{(1)} = -\sum_{\mathbf{q}} \mathbf{j}_e(-\mathbf{q}) \cdot \mathbf{A}(\mathbf{q}) = \tag{126}$$

$$= \sum_{\mathbf{k},\mathbf{q},\lambda} \frac{e}{\sqrt{2\mathcal{V}\varepsilon_0 \omega_{\mathbf{q},\lambda}}} \psi^\dagger_{\mathbf{k}+\mathbf{q}} \begin{pmatrix} \boldsymbol{\epsilon}_{\mathbf{q}\lambda} \cdot \mathbf{v}_a\left(\mathbf{k}+\frac{\mathbf{q}}{2}\right) & i\left(\boldsymbol{\epsilon}_{\mathbf{q}\lambda} \cdot \mathbf{d}\right)\left(\varepsilon_{a,\mathbf{k}+\mathbf{q}} - \varepsilon_{b,\mathbf{k}}\right) \\ i\left(\boldsymbol{\epsilon}_{\mathbf{q}\lambda} \cdot \mathbf{d}\right)\left(\varepsilon_{b,\mathbf{k}+\mathbf{q}} - \varepsilon_{a,\mathbf{k}}\right) & \boldsymbol{\epsilon}_{\mathbf{q}\lambda} \cdot \mathbf{v}_b\left(\mathbf{k}+\frac{\mathbf{q}}{2}\right) \end{pmatrix} \psi_{\mathbf{k}} \left(a_{\mathbf{q}\lambda} + a^\dagger_{-\mathbf{q}\lambda}\right). \tag{127}$$

Za setup od Curtisa: material 2D v xy ravnini, stoječe EM valovanje v z, vzamemo osnovni mode ($m = 1$), ki prispeva največ, v tem primeru sta dve možni polarizaciji:

$$\boldsymbol{\epsilon}_{\text{TE}}^{(m)}(z) = (\hat{\mathbf{z}} \times \hat{\mathbf{q}}) \sin\left(\frac{m\pi}{L} z\right) \tag{128}$$

$$\boldsymbol{\epsilon}_{\text{TM}}^{(m)}(z) = \frac{\omega_0}{\omega_{\mathbf{q},m}} \hat{\mathbf{q}} \sin\left(\frac{m\pi}{L} z\right) + i \frac{c\, q}{\omega_{\mathbf{q},m}} \hat{\mathbf{z}} \cos\left(\frac{m\pi}{L} z\right) \tag{129}$$

kjer je $\mathbf{q} = q(\cos\varphi, \sin\varphi)$ valovni vektor v ravnini, $\omega_0 = \pi c/L$ in $\omega_{\mathbf{q},m} = \sqrt{m^2\omega_0^2 + c^2 q^2}$ (uporabljam notacijo $\omega_{\mathbf{q}} := \omega_{\mathbf{q},1}$). Na sredini resonatorja, pri $z = L/2$, je $\sin(\pi z/L) = 1$ in $\cos(\pi z/L) = 0$, zato:

$$\boldsymbol{\epsilon}_{\text{TE}}\left(\tfrac{L}{2}\right) = \hat{\mathbf{z}} \times \hat{\mathbf{q}} = \begin{pmatrix} -\sin\varphi \\ \cos\varphi \\ 0 \end{pmatrix} \tag{130}$$

$$\boldsymbol{\epsilon}_{\text{TM}}\left(\tfrac{L}{2}\right) = \frac{\omega_0}{\omega_{\mathbf{q}}} \hat{\mathbf{q}} = \frac{\omega_0}{\omega_{\mathbf{q}}} \begin{pmatrix} \cos\varphi \\ \sin\varphi \\ 0 \end{pmatrix} \tag{131}$$

Iz tega imamo pri $z = L/2$ možne sklopitve glede na model ($\mathbf{d} = d(\cos\theta, \sin\theta, 0)$ v ravnini oz. $\mathbf{d} = d\,\hat{\mathbf{z}}$ izven ravnine):

Način(m)	$\boldsymbol{\epsilon} \cdot \mathbf{v}_\gamma\left(\mathbf{k}+\frac{\mathbf{q}}{2}\right)$	$\boldsymbol{\epsilon} \cdot \mathbf{d}$, $\mathbf{d}$ in plane	$\boldsymbol{\epsilon} \cdot \mathbf{d}$, $\mathbf{d} \parallel \hat{\mathbf{z}}$
TE1	$2t_\gamma\left(-\sin\varphi \sin\left(k_x + \frac{q_x}{2}\right) + \cos\varphi \sin\left(k_y + \frac{q_y}{2}\right)\right)$	$d\sin(\theta - \varphi)$	0
TM1	$\frac{\omega_0}{\omega_{\mathbf{q}}} 2t_\gamma\left(\cos\varphi \sin\left(k_x + \frac{q_x}{2}\right) + \sin\varphi \sin\left(k_y + \frac{q_y}{2}\right)\right)$	$\frac{\omega_0}{\omega_{\mathbf{q}}} d\cos(\theta - \varphi)$	$i\,d\,\frac{c\,q}{\omega_{\mathbf{q}}} \cos\left(\frac{\pi}{L} z\right)$ ($= 0$ pri $z = \frac{L}{2}$)
TM2	0	0	$-i\,d\,\frac{c\,q}{\omega_{\mathbf{q},2}}$

dipolni moment je neničeln za liho-sode kombinacije orbital, tega gledamo samo glede na o.c.:

$$\mathbf{d} = \langle \phi_a | \mathbf{x} | \phi_b \rangle = \int \phi_a^*(\mathbf{x})\, \mathbf{x}\, \phi_b(\mathbf{x})\ \mathrm{d}\mathbf{x}\ . \tag{132}$$

V nadaljevanju sem delal z dipolom in plane ($s - p_x$ tip orbital), ampak to zahteva anioztorpen prekrivalni (hopping) integral, zato ploti z $\mathbf{d} \propto (1, 0, 0)$ niso konsistentni za en model. Dipol znotraj celice v z smeri, se ne sklaplja v $m = 1$ načinu na sredini resonatorja. V tem primeru vzamemo dodatno še TM2 ($m = 2$) (kjer privzamemo TM0 način ni prisoten), vendar za relevantne prehode $\mathbf{q} \to 0$, sklopitev s tem načinom izgine.

2.5.1 Model prostih elektronov in sklopitev s fotoni

Kaj pa če gledamo namesto tight binding, proste elektrone, torej

$$\varepsilon_{\gamma\mathbf{k}} = \pm\frac{G}{2} \mp 2t(\cos k_x + \cos k_y) \quad \text{za} \quad k_x, k_y \sim 0 \tag{133}$$

od tod

$$\varepsilon_{\gamma\mathbf{k}} \approx \pm\frac{G}{2} \mp 2t\left(1 - \frac{k_x^2}{2} + 1 - \frac{k_y^2}{2}\right) = \pm\left(\frac{G}{2} - 4t\right) \pm t(k_x^2 + k_y^2) = \pm(\varepsilon^0 + t\mathbf{k} \cdot \mathbf{k}) \tag{134}$$

Torej je Hamiltonka

$$H_e^{(0)} = \sum_{\gamma,\mathbf{k}} \left(\varepsilon_\gamma^0 + t_\gamma \mathbf{k}^2\right) c^\dagger_{\gamma,\mathbf{k}} c_{\gamma,\mathbf{k}} \tag{135}$$

Na drug način lahko poračunamo tok, ker razvijemo

$$\psi(\mathbf{r}) = \frac{1}{\sqrt{\mathcal{V}}} \sum_{\mathbf{k},\gamma=\{a,b\}} e^{i\mathbf{k}\cdot\mathbf{r}} c_{\gamma,\mathbf{k}}\,. \tag{136}$$

in za tok uporabimo

$$\mathbf{j}_e(\mathbf{r}) = -\frac{e}{2mi}\left(\psi^\dagger(\mathbf{r})\nabla\psi(\mathbf{r}) - \nabla\psi^\dagger(\mathbf{r})\psi(\mathbf{r})\right) = -\frac{e}{2mi}\frac{1}{\mathcal{V}}\sum_{\gamma,\mathbf{k},\mathbf{k}'}\left(c^\dagger_{\gamma\mathbf{k}'}c_{\gamma\mathbf{k}}e^{-i\mathbf{k}'\cdot\mathbf{r}}(i\mathbf{k})e^{i\mathbf{k}\cdot\mathbf{r}} - c^\dagger_{\gamma\mathbf{k}'}c_{\gamma\mathbf{k}}(-i\mathbf{k}')e^{-i\mathbf{k}'\cdot\mathbf{r}}e^{i\mathbf{k}\cdot\mathbf{r}}\right) \tag{137}$$

s FT dobimo

$$\mathbf{j}_e(-\mathbf{q}) = \int \mathrm{d}\mathbf{r}\; \mathbf{j}_e(\mathbf{r})e^{i\mathbf{q}\cdot\mathbf{r}} = -\frac{e}{2m}\sum_{\gamma,\mathbf{k}}\left(c^\dagger_{\gamma,\mathbf{k}+\mathbf{q}}c_{\gamma\mathbf{k}}\,\mathbf{k} + (\mathbf{k}+\mathbf{q})c^\dagger_{\gamma,\mathbf{k}+\mathbf{q}}c_{\gamma\mathbf{k}}\right) = -\frac{e}{m}\sum_{\mathbf{k}}\left(\mathbf{k}+\frac{\mathbf{q}}{2}\right)\psi^\dagger_{\mathbf{k}+\mathbf{q}}\psi_{\mathbf{k}} \tag{138}$$

kar bi dobili tudi iz bolj splošne relacije. V tem primeru dobimo enako kot Curtis z $m^* = 1/2t$, oziroma $m^* = \hbar^2/(2ta_0^2)$

$$H^{(1)}_{e\gamma} = \frac{e}{m}\sum_{\mathbf{k},\mathbf{q},\lambda}\frac{1}{\sqrt{2\mathcal{V}\epsilon_0\omega_{\mathbf{q},\lambda}}}\left(\mathbf{k}+\frac{\mathbf{q}}{2}\right)\cdot\boldsymbol{\epsilon}_{\mathbf{q}\lambda}\,\psi^\dagger_{\mathbf{k}+\mathbf{q}}\psi_{\mathbf{k}}\left(a_{\mathbf{q}\lambda} + a^\dagger_{-\mathbf{q}\lambda}\right) \tag{139}$$

2.5.2 Drugi red sklopitve za t.b.

Zacnemo spet:

$$H^{(0)}_e = \sum_{\gamma,j}\varepsilon^{(0)}_\gamma c^\dagger_{\gamma j}c_{\gamma j} - \sum_{j,\boldsymbol{\delta},\gamma} t_\gamma\left(c^\dagger_{\gamma j}c_{\gamma j+\boldsymbol{\delta}} + \text{h.c.}\right) \tag{140}$$

v prisotnosti polja dobi prekrivalni integral fazo:

$$t_\gamma \to t_\gamma e^{i\theta_{j\boldsymbol{\delta}}} \qquad \theta_{j\boldsymbol{\delta}} = e\int_{\mathbf{R}_j}^{\mathbf{R}_j+\boldsymbol{\delta}}\mathbf{A}(\mathbf{r},t)\cdot d\mathbf{r} \tag{141}$$

od tod

$$H^{(0)}_e[\mathbf{A}] = \sum_{\gamma,j}\varepsilon^{(0)}_\gamma c^\dagger_{\gamma j}c_{\gamma j} - \sum_{j,\boldsymbol{\delta},\gamma} t_\gamma\left(e^{i\theta_{j\boldsymbol{\delta}}}c^\dagger_{\gamma j}c_{\gamma j+\boldsymbol{\delta}} + e^{-i\theta_{j\boldsymbol{\delta}}}c^\dagger_{\gamma j+\boldsymbol{\delta}}c_{\gamma j}\right) \tag{142}$$

od kjer razvijemo do drugega reda, prvi red bi moral biti isti kot gor (razen dipolov)

$$e^{i\theta} = 1 + \underbrace{i\theta}_{\mathcal{O}(e\mathbf{A})}\underbrace{-\frac{1}{2}\theta^2}_{\mathcal{O}(e^2A^2)} \tag{143}$$

vstavimo

$$c_{\gamma j} = \frac{1}{\sqrt{\mathcal{V}}}\sum_{\mathbf{k}} e^{i\mathbf{k}\cdot\mathbf{R}_j}c_{\gamma\mathbf{k}} \tag{144}$$

in približek da se polje ne spremeni velko do najbližjih sosedov

$$\int_{\mathbf{R}_j}^{\mathbf{R}_j+\boldsymbol{\delta}}\mathbf{A}(\mathbf{r},t)\cdot d\mathbf{r} \approx \mathbf{A}\left(\mathbf{R}_j + \frac{\boldsymbol{\delta}}{2}\right)\cdot\boldsymbol{\delta} \qquad (\mathbf{A}\text{ se ne spremeni veliko znotraj celice+midpoint approx.}) \tag{145}$$

računamo

$$H_{e\gamma}^{(2)} = -\sum_{j,\boldsymbol{\delta},\gamma} t_\gamma \left(-\frac{\theta_{j\boldsymbol{\delta}}^2}{2}\right)\left(c_{\gamma j}^\dagger c_{\gamma j+\boldsymbol{\delta}} + c_{\gamma j+\boldsymbol{\delta}}^\dagger c_{\gamma j}\right) = \tag{146}$$

$$= \sum_{j,\boldsymbol{\delta},\gamma} t_\gamma \cdot \frac{e^2}{2} \int_{\mathbf{R}_j}^{\mathbf{R}_j+\boldsymbol{\delta}} \mathbf{A}(\mathbf{r},t)\cdot d\mathbf{r} \int_{\mathbf{R}_j}^{\mathbf{R}_j+\boldsymbol{\delta}} \mathbf{A}(\mathbf{r},t)\cdot d\mathbf{r}\left(c_{\gamma j}^\dagger c_{\gamma j+\boldsymbol{\delta}} + \text{h.c.}\right) = \tag{147}$$

$$= \sum_{j,\boldsymbol{\delta},\gamma} t_\gamma \frac{e^2}{2}\left(\sum_{\mathbf{q},\lambda} \frac{1}{\sqrt{2\varepsilon_0 \mathcal{V}\omega_{\mathbf{q}\lambda}}}\left(\boldsymbol{\epsilon}_{\mathbf{q}\lambda}\cdot\boldsymbol{\delta}\right) e^{i\mathbf{q}\cdot\left(\mathbf{R}_j+\frac{\boldsymbol{\delta}}{2}\right)}\left(a_{\mathbf{q}\lambda} + a_{-\mathbf{q}\lambda}^\dagger\right)\right)\left(\sum_{\mathbf{q}',\lambda'} \frac{1}{\sqrt{2\varepsilon_0 \mathcal{V}\omega_{\mathbf{q}'\lambda'}}}\left(\boldsymbol{\epsilon}_{\mathbf{q}'\lambda'}\cdot\boldsymbol{\delta}\right) e^{i\mathbf{q}'\cdot\left(\mathbf{R}_j+\frac{\boldsymbol{\delta}}{2}\right)}\left(a_{\mathbf{q}'\lambda'} + a_{-\mathbf{q}'\lambda'}^\dagger\right)\right. \tag{148}$$

$$\left.\cdot \sum_{\mathbf{k},\mathbf{k}'} \frac{1}{\mathcal{V}} c_{\gamma\mathbf{k}'}^\dagger c_{\gamma\mathbf{k}} e^{-i(\mathbf{k}'-\mathbf{k})\cdot\mathbf{R}_j}\left(e^{i\mathbf{k}\cdot\boldsymbol{\delta}} + e^{i\mathbf{k}'\cdot\boldsymbol{\delta}}\right)\right) \tag{149}$$

ven pade ohranitev momenta za dvofotonske procese

$$\sum_j \frac{1}{\mathcal{V}} e^{i(\mathbf{q}+\mathbf{q}'-\mathbf{k}'+\mathbf{k})\cdot\mathbf{R}_j} = \delta_{\mathbf{k}',\,\mathbf{k}+\mathbf{q}+\mathbf{q}'} \tag{150}$$

kar da na koncu

$$= \sum_{\substack{\gamma,\mathbf{k},\mathbf{q},\mathbf{q}'\\ \lambda,\lambda',\boldsymbol{\delta}}} t_\gamma \frac{e^2}{2}\frac{1}{2\varepsilon_0\mathcal{V}\sqrt{\omega_{\mathbf{q}\lambda}\omega_{\mathbf{q}'\lambda'}}}\left(\boldsymbol{\epsilon}_{\mathbf{q}\lambda}\cdot\boldsymbol{\delta}\right)\left(\boldsymbol{\epsilon}_{\mathbf{q}'\lambda'}\cdot\boldsymbol{\delta}\right)\left(a_{\mathbf{q}\lambda} + a_{-\mathbf{q}\lambda}^\dagger\right)\left(a_{\mathbf{q}'\lambda'} + a_{-\mathbf{q}'\lambda'}^\dagger\right) c_{\gamma,\mathbf{k}+\mathbf{q}+\mathbf{q}'}^\dagger c_{\gamma\mathbf{k}} \tag{151}$$

$$\cdot\left(e^{i\mathbf{k}\cdot\boldsymbol{\delta}} + e^{-i(\mathbf{k}+\mathbf{q}+\mathbf{q}')\cdot\boldsymbol{\delta}}\right)\cdot e^{i(\mathbf{q}+\mathbf{q}')\cdot\frac{\boldsymbol{\delta}}{2}} \tag{152}$$

Poenostavimo zadnjifaktor v kosinus:

$$H_{e\gamma}^{(2)} = \sum_{\substack{\mathbf{k},\mathbf{q},\mathbf{q}'\\ \lambda,\lambda'}} \psi_{\mathbf{k}+\mathbf{q}+\mathbf{q}'}^\dagger M^{\lambda\lambda'}(\mathbf{k},\mathbf{q},\mathbf{q}')\psi_{\mathbf{k}}\cdot\left(a_{\mathbf{q},\lambda} + a_{-\mathbf{q},\lambda}^\dagger\right)\left(a_{\mathbf{q}',\lambda'} + a_{-\mathbf{q}',\lambda'}^\dagger\right) \tag{153}$$

$$M^{\lambda\lambda'}(\mathbf{k},\mathbf{q},\mathbf{q}') = \sum_{\boldsymbol{\delta}} \frac{e^2}{2\varepsilon_0\mathcal{V}\sqrt{\omega_{\mathbf{q},\lambda}\omega_{\mathbf{q}',\lambda'}}}\left(\boldsymbol{\epsilon}_{\mathbf{q},\lambda}\cdot\boldsymbol{\delta}\right)\left(\boldsymbol{\epsilon}_{\mathbf{q}',\lambda'}\cdot\boldsymbol{\delta}\right)\cos\left(\left(\mathbf{k}+\frac{\mathbf{q}+\mathbf{q}'}{2}\right)\cdot\boldsymbol{\delta}\right)\begin{pmatrix} t_a & 0\\ 0 & t_b\end{pmatrix} \tag{154}$$

Za n.n. v square latticu eksplicitno:

$$M^{\lambda\lambda'}(\mathbf{k},\mathbf{q},\mathbf{q}') = \frac{e^2}{2\varepsilon_0\mathcal{V}\sqrt{\omega_{\mathbf{q},\lambda}\omega_{\mathbf{q}',\lambda'}}}\left[\epsilon_{\mathbf{q},\lambda}^x\epsilon_{\mathbf{q}',\lambda'}^x\cos\left(k_x+\frac{q_x+q_x'}{2}\right) + \epsilon_{\mathbf{q},\lambda}^y\epsilon_{\mathbf{q}',\lambda'}^y\cos\left(k_y+\frac{q_y+q_y'}{2}\right)\right]\begin{pmatrix} t_a & 0\\ 0 & t_b\end{pmatrix}. \tag{155}$$

Razen dipolnih elementov, bi to moralo biti konsistentno s prvim redom. **Kako vlkjučimo dipole v te elemente, da bo konsistentno?**

2.6 Populacijska dinamika

V splošnem sta za obe bozonski sklopitvi Hamiltonjana oblike (v prvem redu):

$$H = \sum_{\mathbf{k},\mathbf{q},\lambda} \psi_{\mathbf{k}+\mathbf{q}}^\dagger M^\lambda(\mathbf{k},\mathbf{q})\psi_{\mathbf{k}}\cdot\left(a_{\mathbf{q},\lambda} + a_{-\mathbf{q},\lambda}^\dagger\right) \tag{156}$$

kar prepišemo v diagonalno bazo za EI s transformacijo (32), za lžje račune definiram v diagonalni bazi:

$$d_{\gamma,\mathbf{k}} = \{\alpha_{\mathbf{k}};\gamma=\alpha,\beta_{\mathbf{k}};\gamma=\beta\}\,. \tag{157}$$

Torej je (indeks veje, oziroma polarizacije λ in vsoto bom spuščal v večini primerov):

$$H = \sum_{\substack{\mathbf{k},\mathbf{q},\lambda \\ \gamma,\gamma'}} d^\dagger_{\gamma,\mathbf{k}+\mathbf{q}} \widetilde{M}_{\gamma,\gamma'}(\mathbf{k},\mathbf{q}) d_{\gamma',\mathbf{k}} \cdot (a_{\mathbf{q},\lambda} + a^\dagger_{-\mathbf{q},\lambda}) \tag{158}$$

$$\widetilde{M}(\mathbf{k},\mathbf{q}) = U^\dagger_{\mathbf{k}+\mathbf{q}} M(\mathbf{k},\mathbf{q}) U_{\mathbf{k}} \tag{159}$$

Izpeljava je z von Neumann enačbo, da dobimo pravilne razširjene resonance, na koncu lahko primejamo s Fermijevim zlatim pravilom. Najprej prepišemo Hamiltonjan v interakcijsko sliko v diagonalni bazi je razvoj

$$H_I(t) = \sum_{\substack{\mathbf{k},\mathbf{q},\lambda \\ \gamma,\gamma'}} d^\dagger_{\gamma,\mathbf{k}+\mathbf{q}} \widetilde{M}_{\gamma,\gamma'}(\mathbf{k},\mathbf{q}) d_{\gamma',\mathbf{k}} \cdot \left(a_{\mathbf{q},\lambda}(t) + a^\dagger_{-\mathbf{q},\lambda}(t)\right) e^{i\Omega t} \tag{160}$$

$$\Omega = E_{\gamma,\mathbf{k}+\mathbf{q}} - E_{\gamma',\mathbf{k}} \tag{161}$$

Sledimo enačbi (55), kjer zamenjamo $t - s = t'$ in vzamemo $\mathrm{Tr}_S(\hat{n}_{\lambda,\mathbf{p}} \bullet)$ na obeh straneh in upoštevamo cikličnost sledi:

$$\partial_t n_{\lambda,\mathbf{p}} = -\int_0^t \mathrm{d}t' \left\langle \left[\left[d^\dagger_{\lambda,\mathbf{p}} d_{\lambda,\mathbf{p}}, H_I(t) \right], H_I(t') \right] \right\rangle \tag{162}$$

(note, $\langle \bullet \rangle = \langle \bullet \rangle_S \langle \bullet \rangle_B$) za notranji komutator rabimo:

$$[d^\dagger_A d_B, d^\dagger_C d_D] = d^\dagger_A \underbrace{\{d_B, d^\dagger_C\}}_{\delta_{BC}} d_D - d^\dagger_C \underbrace{\{d^\dagger_A, d_D\}}_{\delta_{AD}} d_B - \underbrace{\{d^\dagger_A, d^\dagger_C\}}_{0} d_B d_D + d^\dagger_C d^\dagger_A \underbrace{\{d_B, d_D\}}_{0} = d^\dagger_A d_D \delta_{BC} - d^\dagger_C d_B \delta_{AD} \tag{163}$$

oziroma:

$$[d^\dagger_A d_A, d^\dagger_C d_D] = d^\dagger_C d_D (\delta_{AC} - \delta_{AD}) \tag{164}$$

Prvi komutator je tako:

$$[n_{\lambda\mathbf{p}}, H_I(t)] = \sum_{\substack{\mathbf{k},\mathbf{q} \\ \gamma,\gamma'}} \widetilde{M}_{\gamma\gamma'}(\mathbf{k},\mathbf{q}) \overbrace{d^\dagger_{\gamma\mathbf{k}+\mathbf{q}} d_{\gamma'\mathbf{k}}}^{S} e^{i\Omega t} \overbrace{\left(a_{\mathbf{q}\lambda} + a^\dagger_{-\mathbf{q}\lambda}\right)}^{B} \left(\delta_{\lambda\gamma}\delta_{\mathbf{p},\mathbf{k}+\mathbf{q}} - \delta_{\lambda\gamma'}\delta_{\mathbf{p},\mathbf{k}}\right) \tag{165}$$

za zunanji komutator nas zanimajo samo operatorji:

$$\left[\left[n_{\lambda\mathbf{p}}, H_I(t)\right], H_I(t')\right] \sim [SB, S'B'] = SBS'B' - S'B'SB = SS'BB' - S'SB'B \tag{166}$$

oziroma bolj natančno (vzamemo še sled, ki se faktorizira za sistem in kopel)

$$\begin{aligned} \sum_{\substack{\mathbf{k},\mathbf{q},\mathbf{k}',\mathbf{q}' \\ \gamma,\gamma',\omega,\omega'}} & \widetilde{M}_{\gamma\gamma'}(\mathbf{k},\mathbf{q}) \widetilde{M}_{\omega\omega'}(\mathbf{k}',\mathbf{q}') \left(\delta_{\lambda\gamma}\delta_{\mathbf{p},\mathbf{k}+\mathbf{q}} - \delta_{\lambda\gamma'}\delta_{\mathbf{p},\mathbf{k}}\right) \cdot \\ & \cdot \Bigg[\left\langle d^\dagger_{\gamma\mathbf{k}+\mathbf{q}} d_{\gamma'\mathbf{k}} d^\dagger_{\omega\mathbf{k}'+\mathbf{q}'} d_{\omega'\mathbf{k}'} \right\rangle e^{i\Omega t} e^{i\Omega' t'} \left\langle \left(a_{\mathbf{q}\lambda} + a^\dagger_{-\mathbf{q}\lambda}\right)(t) \left(a_{\mathbf{q}'\lambda'} + a^\dagger_{-\mathbf{q}'\lambda'}\right)(t') \right\rangle - \\ & - \left\langle d^\dagger_{\omega\mathbf{k}'+\mathbf{q}'} d_{\omega'\mathbf{k}'} d^\dagger_{\gamma\mathbf{k}+\mathbf{q}} d_{\gamma'\mathbf{k}} \right\rangle e^{i\Omega' t'} e^{i\Omega t} \left\langle \left(a_{\mathbf{q}'\lambda'} + a^\dagger_{-\mathbf{q}'\lambda'}\right)(t') \left(a_{\mathbf{q}\lambda} + a^\dagger_{-\mathbf{q}\lambda}\right)(t) \right\rangle \Bigg], \end{aligned} \tag{167}$$

kjer je $\Omega' = E_{\omega,\mathbf{k}'+\mathbf{q}'} - E_{\omega',\mathbf{k}'}$. Najprej bozonski deli, neničelni so samo mešani produkti (privzel sem še $\omega_{\mathbf{q}} = \omega_{-\mathbf{q}}$):

$$\begin{aligned} \left\langle \left(a_{\mathbf{q},\lambda} + a^\dagger_{-\mathbf{q},\lambda}\right)(t) \left(a_{\mathbf{q}',\lambda'} + a^\dagger_{-\mathbf{q}',\lambda'}\right)(t') \right\rangle &= \left\langle a_{\mathbf{q},\lambda}(t) a^\dagger_{-\mathbf{q}',\lambda'}(t') \right\rangle + \left\langle a^\dagger_{-\mathbf{q},\lambda}(t) a_{\mathbf{q}',\lambda'}(t') \right\rangle = \\ &= \left\langle a_{\mathbf{q},\lambda}(t) a^\dagger_{\mathbf{q},\lambda}(t') \right\rangle + \left\langle a^\dagger_{-\mathbf{q},\lambda}(t) a_{-\mathbf{q},\lambda}(t') \right\rangle = \\ &= (N(\omega_{\mathbf{q}}) + 1) e^{-i\omega_{\mathbf{q}}(t-t')} + N(\omega_{\mathbf{q}}) e^{i\omega_{\mathbf{q}}(t-t')} \end{aligned} \tag{168}$$

kjer mora veljati $\mathbf{q}' = -\mathbf{q}$ in $\lambda = \lambda'$ (note, ta lambda ni isti kot un fermionski lambda, zmanjkuje mi lepih grskih crk za pisat). Podobno je z drugim členom. Uvedemo še $\tau = t - t'$, integral se ne spremeni z uvedbo te spremenljivke. A la Wick naredimo še za fermionske dele (ker so d diagonalni lahko uporabimo Wicka):

$$\langle d_A^\dagger d_B d_C^\dagger d_D \rangle = \langle d_A^\dagger d_B \rangle \langle d_C^\dagger d_D \rangle + \langle d_A^\dagger d_D \rangle \langle d_B d_C^\dagger \rangle = \tag{169}$$

$$= n_A n_C \delta_{AB} \delta_{CD} + n_A (1 - n_C) \delta_{BC} \delta_{AD} \tag{170}$$

$$\hookrightarrow \delta_{AB} \mapsto \delta_{\gamma\gamma'} \delta_{\mathbf{k}+\mathbf{q},\mathbf{k}} = 0 \quad \text{odpade zaradi razlike delta funkcij} \tag{171}$$

oziroma (prvi člen se izvrednoti v nič, zaradi razlike delta funckij, sicer pa to predstavlja ničti red sipanja), upoštevamo še $\mathbf{q}' = -\mathbf{q}$:

$$\left\langle d^\dagger_{\gamma\mathbf{k}+\mathbf{q}} d_{\gamma'\mathbf{k}} d^\dagger_{\omega\mathbf{k}'+\mathbf{q}'} d_{\omega'\mathbf{k}'} \right\rangle = n_{\gamma,\mathbf{k}+\mathbf{q}} \left(1 - n_{\omega,\mathbf{k}'-\mathbf{q}}\right) \delta_{\gamma'\omega} \delta_{\mathbf{k},\mathbf{k}'-\mathbf{q}} \delta_{\gamma\omega'} \tag{172}$$

$$\left\langle d^\dagger_{\omega\mathbf{k}'+\mathbf{q}'} d_{\omega'\mathbf{k}'} d^\dagger_{\gamma\mathbf{k}+\mathbf{q}} d_{\gamma'\mathbf{k}} \right\rangle = n_{\omega,\mathbf{k}'-\mathbf{q}} \left(1 - n_{\gamma,\mathbf{k}+\mathbf{q}}\right) \delta_{\omega'\gamma} \delta_{\mathbf{k}',\mathbf{k}+\mathbf{q}} \delta_{\omega\gamma'} \tag{173}$$

iz delt sledi za eksponenta/časovni razvoj v obeh clenih:

$$\Omega' = E_{\omega,\mathbf{k}'+\mathbf{q}'} - E_{\omega',\mathbf{k}'} = E_{\gamma',\mathbf{k}} - E_{\gamma,\mathbf{k}+\mathbf{q}} = -\Omega \implies e^{i\Omega' t'} e^{i\Omega t} = e^{i\Omega\tau} \tag{174}$$

Vse skupaj je treba še integrirati, dodamo regulator, da pole izmaknemo iz osi:

$$\int_0^\infty e^{i\tau(\Omega \pm \omega_\mathbf{q} + i\gamma)} \, d\tau = -\frac{i e^{i(\Omega \pm \omega_\mathbf{q})\tau} e^{-\gamma\tau}}{\Omega \pm \omega_\mathbf{q} + i\gamma} \bigg|_0^\infty = \frac{i}{\Omega \pm \omega_\mathbf{q} + i\gamma} = D_\pm(\Omega) \tag{175}$$

oziroma skupaj:

$$\partial_t n_{\lambda,\mathbf{p}} = -\int_0^{t\to\infty} \sum_{\substack{\mathbf{k},\mathbf{q} \\ \gamma,\gamma'}} \left(\delta_{\lambda\gamma}\delta_{\mathbf{p},\mathbf{k}+\mathbf{q}} - \delta_{\lambda\gamma'}\delta_{\mathbf{p},\mathbf{k}}\right) \cdot \tag{176}$$

$$\cdot \Big(n_{\gamma,\mathbf{k}+\mathbf{q}} (1 - n_{\gamma',\mathbf{k}}) \widetilde{M}_{\gamma\gamma'}(\mathbf{k},\mathbf{q}) \widetilde{M}_{\gamma'\gamma}(\mathbf{k}+\mathbf{q},-\mathbf{q}) e^{i\Omega\tau} \left((N(\omega_\mathbf{q}) + 1) e^{-i\omega_\mathbf{q}\tau} + N(\omega_\mathbf{q}) e^{i\omega_\mathbf{q}\tau} \right) \tag{177}$$

$$- n_{\gamma',\mathbf{k}} (1 - n_{\gamma,\mathbf{k}+\mathbf{q}}) \widetilde{M}_{\gamma\gamma'}(\mathbf{k},\mathbf{q}) \widetilde{M}_{\gamma'\gamma}(\mathbf{k}+\mathbf{q},-\mathbf{q}) e^{i\Omega\tau} \left((N(\omega_\mathbf{q}) + 1) e^{i\omega_\mathbf{q}\tau} + N(\omega_\mathbf{q}) e^{-i\omega_\mathbf{q}\tau} \right) \Big) \, \mathrm{d}\tau \tag{178}$$

Iz hermitskosti sledi:

$$\widetilde{M}_{\gamma'\gamma}(\mathbf{k}+\mathbf{q},-\mathbf{q}) = \widetilde{M}_{\gamma'\gamma}(\mathbf{k},\mathbf{q})^* \tag{179}$$

ko razpišemo vseh 8 členov z deltami in preimenujemo indekse ostane ($\Omega = E_{\lambda,\mathbf{p}} - E_{\gamma,\mathbf{k}}$)

$$\partial_t n_{\lambda,\mathbf{p}} = -\sum_{\mathbf{k},\gamma} \left| \widetilde{M}_{\gamma\lambda} \right|^2 n_{\lambda\mathbf{p}} (1 - n_{\gamma\mathbf{k}}) \left\{ (N_{\mathbf{p}-\mathbf{k}} + 1) \left[D_-(\Omega) + D_+(-\Omega) \right] + N_{\mathbf{p}-\mathbf{k}} \left[D_+(\Omega) + D_-(-\Omega) \right] \right\} + \tag{180}$$

$$+ \sum_{\mathbf{k},\gamma} \left| \widetilde{M}_{\lambda\gamma} \right|^2 (1 - n_{\lambda\mathbf{p}}) n_{\gamma\mathbf{k}} \left\{ (N_{\mathbf{p}-\mathbf{k}} + 1) \left[D_+(\Omega) + D_-(-\Omega) \right] + N_{\mathbf{p}-\mathbf{k}} \left[D_-(\Omega) + D_+(-\Omega) \right] \right\} \tag{181}$$

od kjer lahko poenostavimo:

$$D_-(\Omega) + D_+(-\Omega) = \frac{2\gamma}{(\Omega - \omega_\mathbf{q})^2 + \gamma^2} \equiv \mathcal{A}_\mathbf{q}(\Omega) \tag{182}$$

$$D_+(\Omega) + D_-(-\Omega) = \frac{2\gamma}{(\Omega + \omega_\mathbf{q})^2 + \gamma^2} = \mathcal{A}_\mathbf{q}(-\Omega) \tag{183}$$

oziroma končno:;

$$\partial_t n_{\gamma\mathbf{k}} = \sum_{\mathbf{k}',\gamma'} \Gamma_{\gamma'\mathbf{k}'\to\gamma\mathbf{k}} n_{\gamma'\mathbf{k}'} (1 - n_{\gamma\mathbf{k}}) - \Gamma_{\gamma\mathbf{k}\to\gamma'\mathbf{k}'} n_{\gamma\mathbf{k}} (1 - n_{\gamma'\mathbf{k}'}) \tag{184}$$

$$\Gamma_{\gamma'\mathbf{k}'\to\gamma\mathbf{k}} = \sum_\lambda \left| \tilde{M}^\lambda_{\gamma\gamma'}(\mathbf{k}', \mathbf{k} - \mathbf{k}') \right|^2 \cdot \left\{ (N_\lambda(-\Omega) + 1) \mathcal{A}_{\mathbf{k}-\mathbf{k}',\lambda}(-\Omega) + N_\lambda(\Omega) \mathcal{A}_{\mathbf{k}-\mathbf{k}',\lambda}(\Omega) \right\} \tag{185}$$

$$\Gamma_{\gamma\mathbf{k}\to\gamma'\mathbf{k}'} = \sum_\lambda \left| \tilde{M}^\lambda_{\gamma\gamma'}(\mathbf{k}', \mathbf{k} - \mathbf{k}') \right|^2 \cdot \left\{ (N_\lambda(\Omega) + 1) \mathcal{A}_{\mathbf{k}'-\mathbf{k},\lambda}(\Omega) + N_\lambda(-\Omega) \mathcal{A}_{\mathbf{k}'-\mathbf{k},\lambda}(-\Omega) \right\} \tag{186}$$

$$\Omega = E_{\gamma\mathbf{k}} - E_{\gamma'\mathbf{k}'}, \qquad N(-\Omega) = -1 - N(\Omega) \tag{187}$$

preverimo predznake v tabeli 1.

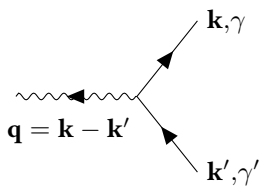
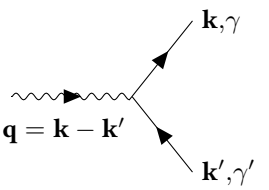
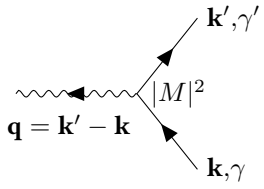
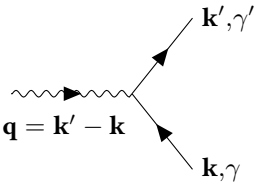
A1  $\Omega+\omega_{\mathbf{q}}\simeq 0$ $= E_{\gamma\mathbf{k}} - E_{\gamma'\mathbf{k}'} + \omega_{\mathbf{q}}$ $= \underbrace{(E_{\gamma\mathbf{k}}+\omega_{\mathbf{q}})}_{E_f} - \underbrace{E_{\gamma'\mathbf{k}'}}_{E_i}.$	A2  $\Omega-\omega_{\mathbf{q}}\simeq 0$ $= E_{\gamma\mathbf{k}} - E_{\gamma'\mathbf{k}'} - \omega_{\mathbf{q}}$ $= \underbrace{E_{\gamma\mathbf{k}}}_{E_f} - \underbrace{(E_{\gamma'\mathbf{k}'}+\omega_{\mathbf{q}})}_{E_i}.$
B1  $\Omega-\omega_{\mathbf{q}}\simeq 0$ $= E_{\gamma\mathbf{k}} - E_{\gamma'\mathbf{k}'} - \omega_{\mathbf{q}}$ $= -\big[(E_{\gamma'\mathbf{k}'}+\omega_{\mathbf{q}}) - E_{\gamma\mathbf{k}}\big].$	B2  $\Omega+\omega_{\mathbf{q}}\simeq 0$ $= E_{\gamma\mathbf{k}} - E_{\gamma'\mathbf{k}'} + \omega_{\mathbf{q}}$ $= -\big[E_{\gamma'\mathbf{k}'} - (E_{\gamma\mathbf{k}}+\omega_{\mathbf{q}})\big].$

Tabela 1 | Feynman diagrami za vse štiri procese (iz in v fiskno stanje $\gamma, \mathbf{k}$).

Dobimo enako kot Curtis, ampak smo za fotone privzeli, da cavity ni leaky; za korelatorje oziroma produkte bozonov smo privzeli da imajo točno definirane resonance in dodali regulator na roke. V resnici so fotoni sklopljeni z okolico, kar jim da končno širino v disperziji. Rate matrike lahko še poenostavimo na razlike Lorentzovih vrhov:

$$\Gamma_{\gamma'\mathbf{k}'\to\gamma\mathbf{k}} = |\widetilde{M}_{\gamma\gamma'}(\mathbf{k}', \mathbf{k}-\mathbf{k}')|^2 N(\Omega)\cdot\Big\{\mathcal{A}_{\mathbf{k}-\mathbf{k}'}(\Omega) - \mathcal{A}_{\mathbf{k}-\mathbf{k}'}(-\Omega)\Big\} \tag{188}$$

$$\Gamma_{\gamma\mathbf{k}\to\gamma'\mathbf{k}'} = |\widetilde{M}_{\gamma\gamma'}(\mathbf{k}', \mathbf{k}-\mathbf{k}')|^2 (N(\Omega)+1)\cdot\Big\{\mathcal{A}_{\mathbf{k}'-\mathbf{k}}(\Omega) - \mathcal{A}_{\mathbf{k}'-\mathbf{k}}(-\Omega)\Big\} \tag{189}$$

od kjer vidimo, da s tem točno zadostimo detaljnemu ravnovesju(?nevem kaj je prevod):

$$\frac{\Gamma_{\gamma'\mathbf{k}'\to\gamma\mathbf{k}}}{\Gamma_{\gamma\mathbf{k}\to\gamma'\mathbf{k}'}} = \frac{N(\Omega)}{N(\Omega)+1} = e^{-\Omega/T}. \tag{190}$$

Če vzamemo GGE obliko, za boljšo stabilnost: $n_{\gamma\mathbf{k}} = 1/(1+\exp(\mu_{\gamma\mathbf{k}}))$, potem je

$$\partial_t n_{\gamma\mathbf{k}} = -n_{\gamma\mathbf{k}}(1-n_{\gamma\mathbf{k}})\partial_t\mu_{\gamma\mathbf{k}} = \mathcal{I}_{\gamma\mathbf{k}} \tag{191}$$

$$\mathcal{I}_{\gamma\mathbf{k}} = \sum_{\gamma',\mathbf{k}'}\left[\Gamma_{\gamma'\mathbf{k}'\to\gamma\mathbf{k}}(1-n_{\gamma\mathbf{k}})n_{\gamma'\mathbf{k}'} - \Gamma_{\gamma\mathbf{k}\to\gamma'\mathbf{k}'}(1-n_{\gamma'\mathbf{k}'})n_{\gamma\mathbf{k}}\right] \tag{192}$$

$$\partial_t\mu_{\gamma\mathbf{k}} = -\frac{\mathcal{I}_{\gamma\mathbf{k}}}{n_{\gamma\mathbf{k}}(1-n_{\gamma\mathbf{k}})} \tag{193}$$

oziroma:

$$\partial_t \mu_{\gamma\mathbf{k}} = -\sum_{\gamma',\mathbf{k}',b} \Gamma^b_{\gamma'\mathbf{k}'\to\gamma\mathbf{k}} \frac{1+e^{\mu_{\gamma\mathbf{k}}}}{1+e^{\mu_{\gamma'\mathbf{k}'}}} - \Gamma^b_{\gamma\mathbf{k}\to\gamma'\mathbf{k}'} \frac{1+e^{-\mu_{\gamma\mathbf{k}}}}{1+e^{-\mu_{\gamma'\mathbf{k}'}}}\,, \tag{194}$$

kjer je $\Gamma^b = \Gamma(T_b)$. Izraz se poenostavi če upoštevamo še zahtevo po d.b.:

$$\partial_t \mu_{\gamma\mathbf{k}} = \sum_{\gamma'\mathbf{k}',b} \Gamma^b_{\gamma'\mathbf{k}'\to\gamma\mathbf{k}} \frac{1+e^{\mu_{\gamma\mathbf{k}}}}{1+e^{\mu_{\gamma'\mathbf{k}'}}} \left(e^{\Omega/T_b + \mu_{\gamma'\mathbf{k}'} - \mu_{\gamma\mathbf{k}}} - 1\right) \tag{195}$$

2.7 Sprememba koordinat

Za 2D square lattice mrežo uvedemo:

$$s = \cos k_x + \cos k_y, \qquad u = \cos k_x - \cos k_y. \tag{196}$$

Obratna transformacija je bijektivna na irreducible trikotniku ($\Gamma - X - M$, oziroma: $\{(k_x,k_y) : 0 \le k_y \le k_x \le \pi\}$):

$$\mathbf{k}(s,u) = \left(\arccos\left(\frac{s+u}{2}\right), \arccos\left(\frac{s-u}{2}\right)\right)^{\mathsf{T}} \tag{197}$$

Z nekaj obračanja dobimo območje kjer sta definirani novi spremenljivki:

$$-2 \le s \le 2, \qquad |s| - 2 \le u \le 0 \tag{198}$$

$$\varepsilon_a(s) = \frac{G}{2} - 2t_a s, \qquad \varepsilon_b(s) = -\frac{G}{2} - 2t_b s. \tag{199}$$

$$\varepsilon^H_a(s) = \frac{G}{2} - 2t_a s + \frac{V}{2}(n-m), \tag{200}$$

$$\varepsilon^H_b(s) = -\frac{G}{2} - 2t_b s + \frac{V}{2}(n+m). \tag{201}$$

$$\eta(s) = \frac{\varepsilon^H_a(s) + \varepsilon^H_b(s)}{2} = \frac{Vn}{2} - (t_a + t_b)s, \tag{202}$$

$$\xi(s) = \frac{\varepsilon^H_a(s) - \varepsilon^H_b(s)}{2} = \frac{G - Vm}{2} - (t_a - t_b)s. \tag{203}$$

$$E_{\alpha,\beta}(s,u) = \frac{Vn}{2} - (t_a + t_b)s \;\pm\; \sqrt{\left[\frac{G-Vm}{2} - (t_a - t_b)s\right]^2 + |\Delta|^2}. \tag{204}$$

$$u_{\mathbf{k}} \to u^2(s) = \frac{1}{2}\left(1 + \frac{\xi(s)}{\sqrt{\xi^2(s) + \Delta^2}}\right), \tag{205}$$

$$v_{\mathbf{k}} \to v^2(s) = \frac{1}{2}\left(1 - \frac{\xi(s)}{\sqrt{\xi^2(s) + \Delta^2}}\right), \tag{206}$$

$$u_{\mathbf{k}} v_{\mathbf{k}} \to u(s)v(s) = -\frac{\Delta}{2\sqrt{\xi^2(s) + \Delta^2}}. \tag{207}$$

Za integrale/fazne vsote po 1. B.C. je Jakobijan

$$\left|\frac{\partial(s,u)}{\partial(k_x,k_y)}\right| = 2\sin k_x \sin k_y. \tag{208}$$

Za intergal po 1. B.C. lahko v najbolj simetričnem primeru (za funckijo $F(g\mathbf{k}) = F(\mathbf{k})$,$\forall g \in C_{4v}$) pišemo (upoštevamo integral po trikotniku $\cdot 8$)

$$\langle F \rangle_{\text{B.C.}} = \frac{1}{(2\pi)^2} \int_{\text{B.C.}} d^2k\, F(\mathbf{k}) = \frac{1}{\pi^2} \int_{-2}^{2} ds \int_{|s|-2}^{0} du\, \frac{F(s,u)}{\sqrt{1-\frac{(s+u)^2}{4}}\sqrt{1-\frac{(s-u)^2}{4}}}. \tag{209}$$

Velikokrat je $F(s,u) = F(s)$, takrat velja:

$$\pi^2 g(s) = \int_{|s|-2}^{0} du \frac{1}{\sqrt{1-\frac{(s+u)^2}{4}}\sqrt{1-\frac{(s-u)^2}{4}}} = \frac{4}{2+|s|} K\left(\frac{2-|s|}{2+|s|}\right) \tag{210}$$

$$K(k) = \int_0^{\pi/2} \frac{d\theta}{\sqrt{1-k^2\sin^2\theta}} \tag{211}$$

okoli $s \approx 0$:

$$\pi^2 g(s) \approx (2-|s|)\cdot\frac{1}{2}\ln\left(\frac{8}{|s|}\right) \approx \ln\left(\frac{8}{|s|}\right) \tag{212}$$

$$\langle F \rangle_{\text{B.C.}} = \int_{-2}^{2} g(s)\cdot F(s)\, ds \quad ; \quad g(s) = \frac{1}{\pi^2}\frac{4}{2+|s|} K\left(\frac{2-|s|}{2+|s|}\right) \tag{213}$$

V termičnem primeru velja:

$$\Delta = V \left\langle \frac{\Delta}{2\sqrt{\xi^2(s)+\Delta^2}} [f_\beta(s) - f_\alpha(s)] \right\rangle_{\text{B.C.}}, \tag{214}$$

$$m = \left\langle \frac{\xi(s)}{\sqrt{\xi^2(s)+\Delta^2}} [f_\alpha(s) - f_\beta(s)] \right\rangle_{\text{B.C.}}, \tag{215}$$

$$n = \langle f_\alpha(s) + f_\beta(s) \rangle_{\text{B.C.}}. \tag{216}$$

Za najbolj splošen primer je integral enak:

$$\langle F(\mathbf{k}) \rangle_{\text{B.C.}} = = \frac{1}{8\pi^2} \sum_{g\in C_{4v}} \int_{-2}^{2} ds \int_{|s|-2}^{0} du\, \frac{F(g\mathbf{k}(s,u))}{\sqrt{1-\frac{(s+u)^2}{4}}\sqrt{1-\frac{(s-u)^2}{4}}}. \tag{217}$$

Najprej opazimo da je za $g \in C_{4v}$ spremeljivka s in vse količine, ki so samo funkcije s invariantna, spremeljivka u pa se transfromira kot $u \to u\,, g = \{E, C_2, \sigma_v, \sigma_{v'}\}$ in $u \to -u$ za ostale 4 elemente. Torej, če je $F(s,u)$ soda v u je invariantna, v splošnem pa je:

$$\sum_{g\in C_{4v}} F(g\mathbf{k}) = 4F(s,u) + 4F(s,-u)\,. \tag{218}$$

zato je v splošnem:

$$\langle F(\mathbf{k}) \rangle_{\text{B.C.}} = \frac{1}{2\pi^2} \int_{-2}^{2} ds \int_{|s|-2}^{0} du\, \frac{F(s,u)+F(s,-u)}{\sqrt{1-\frac{(s+u)^2}{4}}\sqrt{1-\frac{(s-u)^2}{4}}}. \tag{219}$$

Za netermični primer je rahla komplikacija z $f_\gamma(s) \mapsto n_\gamma(s,u)$. Prepisati moramo sipalni integral $\mathcal{I}_{\gamma\mathbf{k}}$ na nove spremenljivke.

$$\partial_t \mu_{\gamma\mathbf{k}} = \sum_{\gamma',b} \frac{1}{2\pi^2} \int_{-2}^{2} ds' \int_{|s|-2}^{0} du'\, J(s',u') \sum_{i=1}^{2} \left[\Gamma^b_{\gamma'\mathbf{k}'_i \to \gamma\mathbf{k}} \frac{1+e^{\mu_{\gamma\mathbf{k}}}}{1+e^{\mu_{\gamma'\mathbf{k}'_i}}} \left(e^{\Omega/T_b + \mu_{\gamma'\mathbf{k}'_i} - \mu_{\gamma\mathbf{k}}} - 1 \right) \right] \tag{220}$$

$$\Gamma^b_{\gamma'\mathbf{k}'_i \to \gamma\mathbf{k}} = |\widetilde{M}_{\gamma\gamma'}(\mathbf{k}'_i, \mathbf{k}-\mathbf{k}'_i)|^2 N(\Omega) \cdot \left\{ \mathcal{A}_{\mathbf{k}-\mathbf{k}'_i}(\Omega) - \mathcal{A}_{\mathbf{k}-\mathbf{k}'_i}(-\Omega) \right\} \tag{221}$$

$$\mathbf{k}'_1 = \mathbf{k}'(s',u')\,, \quad \mathbf{k}'_2 = \mathbf{k}'(s',-u') \tag{222}$$

kjer

$$\mathbf{k}'_1 = (k'_{x1},\, k'_{y1}) = \left(\arccos\left(\frac{s'+u'}{2}\right),\, \arccos\left(\frac{s'-u'}{2}\right)\right) \tag{223}$$

$$\mathbf{k}'_2 = (k'_{x2},\, k'_{y2}) = \left(\arccos\left(\frac{s'-u'}{2}\right),\, \arccos\left(\frac{s'+u'}{2}\right)\right) \tag{224}$$

če je Γ popolnoma simetrična na vse operacije vsaj vemo, da je $\mu_\gamma(s,u) = \mu_\gamma(s,-u)$, ne določa pa to odvisnosti od u. Recimo, da fiksiramo $\omega_{\mathbf{q}} = \sqrt{\omega_0^2 + c^2q^2}$. Ta je invariantna na C_{4v}, tudi $\Omega = \Omega(s)$ je, zato je $\mathcal{A}_{\mathbf{q}}(\Omega)$ invariantna. Enako velja za Bose faktorje. Za dipolne elemente v z smeri, je Γ maksimalno simetrična, ampak ima še vseeno odvisnost od koordinate u.

3 Konkretna implementacija za sklopitev s fotoni

konkretna implementacija/racun za verteks za fotone:

$$\mathbf{q} = q(\cos\varphi, \sin\varphi), \qquad \varphi = \text{atan2}(q_y,q_x)\ (\varphi = 0 \text{ pri } \mathbf{q} = 0), \qquad \omega_{\mathbf{q}} = \sqrt{\omega_0^2 + c^2q^2}, \tag{225}$$

$$s_{\mathbf{k}} = \cos k_x + \cos k_y, \qquad \varepsilon_{a,\mathbf{k}} = \tfrac{G}{2} - 2t_a s_{\mathbf{k}}, \qquad \varepsilon_{b,\mathbf{k}} = -\tfrac{G}{2} - 2t_b s_{\mathbf{k}}. \tag{226}$$

$$\mathbf{v}_\gamma(\mathbf{p}) = \nabla_{\mathbf{p}}\, \varepsilon_{\gamma,\mathbf{p}} = \nabla_{\mathbf{p}} \left[\pm\tfrac{G}{2} - 2t_\gamma(\cos p_x + \cos p_y)\right] = 2t_\gamma\, (\sin p_x,\ \sin p_y). \tag{227}$$

$$\mathbf{v}_\gamma\big(\mathbf{k}' + \tfrac{\mathbf{q}}{2}\big) = 2t_\gamma\, (\sin\big(k'_x + \tfrac{q_x}{2}\big)\,,\ \sin\big(k'_y + \tfrac{q_y}{2}\big)), \tag{228}$$

$$\boldsymbol{\epsilon}_\lambda \cdot \mathbf{v}_\gamma\big(\mathbf{k}' + \tfrac{\mathbf{q}}{2}\big) = 2t_\gamma\, X_\lambda, \qquad X_\lambda = \boldsymbol{\epsilon}_\lambda \cdot (\sin\big(k'_x + \tfrac{q_x}{2}\big)\,,\ \sin\big(k'_y + \tfrac{q_y}{2}\big)). \tag{229}$$

$$\boldsymbol{\epsilon}_{\text{TE}} = (-\sin\varphi,\ \cos\varphi), \qquad \boldsymbol{\epsilon}_{\text{TM}} = \frac{\omega_0}{\omega_{\mathbf{q}}}(\cos\varphi,\ \sin\varphi), \qquad \mathbf{d} = d\,(1,0). \tag{230}$$

$$X_{\text{TE}} = -\sin\varphi\, h_x + \cos\varphi\, h_y, \qquad X_{\text{TM}} = \frac{\omega_0}{\omega_{\mathbf{q}}}\,(\cos\varphi\, h_x + \sin\varphi\, h_y)\,, \tag{231}$$

$$P_{\text{TE}} = -d\sin\varphi, \qquad P_{\text{TM}} = \frac{\omega_0}{\omega_{\mathbf{q}}}\, d\cos\varphi, \tag{232}$$

$$\Delta_{ab} = \varepsilon_{a,\mathbf{k}} - \varepsilon_{b,\mathbf{k}'} = G - 2t_a s_{\mathbf{k}} + 2t_b s_{\mathbf{k}'}, \qquad \Delta_{ba} = \varepsilon_{b,\mathbf{k}} - \varepsilon_{a,\mathbf{k}'} = -G - 2t_b s_{\mathbf{k}} + 2t_a s_{\mathbf{k}'}. \tag{233}$$

$$M^\lambda(\mathbf{k}',\mathbf{q}) = \sqrt{\frac{A}{\omega_{\mathbf{q}}}} \begin{pmatrix} 2t_a X_\lambda & iP_\lambda \Delta_{ab} \\ iP_\lambda \Delta_{ba} & 2t_b X_\lambda \end{pmatrix} \tag{234}$$

$$\widetilde{M}^\lambda = U_{\mathbf{k}}^T M^\lambda U_{\mathbf{k}'}, \qquad U_{\mathbf{k}} = \begin{pmatrix} u & v \\ -v & u \end{pmatrix}, \qquad u = \sqrt{\tfrac{1}{2}\left(1 + \tfrac{\xi_{\mathbf{k}}}{E_{\mathbf{k}}}\right)}, \quad v = \text{sgn}(\Delta)\sqrt{\tfrac{1}{2}\left(1 - \tfrac{\xi_{\mathbf{k}}}{E_{\mathbf{k}}}\right)}. \tag{235}$$

$$\widetilde{M}^\lambda_{\alpha\alpha} = \sqrt{\frac{A}{\omega_{\mathbf{q}}}}\, [X_\lambda\, (2t_a uu' + 2t_b vv') - iP_\lambda\, (uv'\Delta_{ab} + vu'\Delta_{ba})]\,, \tag{236}$$

$$\widetilde{M}^\lambda_{\alpha\beta} = \sqrt{\frac{A}{\omega_{\mathbf{q}}}}\, [X_\lambda\, (2t_a uv' - 2t_b vu') + iP_\lambda\, (uu'\Delta_{ab} - vv'\Delta_{ba})]\,, \tag{237}$$

$$\widetilde{M}^\lambda_{\beta\alpha} = \sqrt{\frac{A}{\omega_{\mathbf{q}}}}\, [X_\lambda\, (2t_a vu' - 2t_b uv') + iP_\lambda\, (uu'\Delta_{ba} - vv'\Delta_{ab})]\,, \tag{238}$$

$$\widetilde{M}^\lambda_{\beta\beta} = \sqrt{\frac{A}{\omega_{\mathbf{q}}}}\, [X_\lambda\, (2t_a vv' + 2t_b uu') + iP_\lambda\, (vu'\Delta_{ab} + uv'\Delta_{ba})]\,. \tag{239}$$

$$\sum_{\lambda}\left|\widetilde{M}^{\lambda}_{\gamma\gamma'}\right|^2=\frac{A}{\omega_{\mathbf{q}}}\left[\left(X^2_{\mathrm{TE}}+X^2_{\mathrm{TM}}\right)C^2_{\gamma\gamma'}+\left(\sin^2\varphi+\frac{\omega_0^2}{\omega_{\mathbf{q}}^2}\cos^2\varphi\right)D^2_{\gamma\gamma'}\right], \tag{240}$$

$$C_{\alpha\alpha}=2t_a uu'+2t_b vv', \qquad D_{\alpha\alpha}=d\,(uv'\Delta_{ab}+vu'\Delta_{ba}), \tag{241}$$

$$C_{\alpha\beta}=2t_a uv'-2t_b vu', \qquad D_{\alpha\beta}=d\,(uu'\Delta_{ab}-vv'\Delta_{ba}), \tag{242}$$

$$C_{\beta\alpha}=2t_a vu'-2t_b uv', \qquad D_{\beta\alpha}=d\,(uu'\Delta_{ba}-vv'\Delta_{ab}), \tag{243}$$

$$C_{\beta\beta}=2t_a vv'+2t_b uu', \qquad D_{\beta\beta}=d\,(vu'\Delta_{ab}+uv'\Delta_{ba}). \tag{244}$$

$$\begin{aligned}\Gamma_{\gamma'\mathbf{k}'\to\gamma\mathbf{k}}&=\sum_{\lambda}\left|\widetilde{M}^{\lambda}_{\gamma\gamma'}\right|^2 N(\Omega)\,\frac{8\gamma\,\omega_{\mathbf{q}}\,\Omega}{\left[(\Omega-\omega_{\mathbf{q}})^2+\gamma^2\right]\left[(\Omega+\omega_{\mathbf{q}})^2+\gamma^2\right]}=\\&=A\Bigg[\Bigg(\left(\sin^2\varphi+\frac{\omega_0^2}{\omega_{\mathbf{q}}^2}\cos^2\varphi\right)\sin^2\left(k'_x+\tfrac{q_x}{2}\right)+\left(\cos^2\varphi+\frac{\omega_0^2}{\omega_{\mathbf{q}}^2}\sin^2\varphi\right)\sin^2\left(k'_y+\tfrac{q_y}{2}\right)+\\&+2\left(\frac{\omega_0^2}{\omega_{\mathbf{q}}^2}-1\right)\cos(\varphi)\sin(\varphi)\sin\left(k'_x+\tfrac{q_x}{2}\right)\sin\left(k'_y+\tfrac{q_y}{2}\right)\Bigg)C^2_{\gamma\gamma'}+\\&+\left(\sin^2\varphi+\frac{\omega_0^2}{\omega_{\mathbf{q}}^2}\cos^2\varphi\right)D^2_{\gamma\gamma'}\Bigg]\cdot\Omega N(\Omega)\cdot\frac{8\gamma}{\left[(\Omega-\omega_{\mathbf{q}})^2+\gamma^2\right]\left[(\Omega+\omega_{\mathbf{q}})^2+\gamma^2\right]}.\end{aligned} \tag{245}$$

Za dipol v z smeri, obdržimo še TM2 mode, čeprav izgine za $q\to 0$:

$$\boldsymbol{\epsilon}_{\mathrm{TE1}}=(-\sin\varphi,\ \cos\varphi,\ 0), \qquad \boldsymbol{\epsilon}_{\mathrm{TM1}}=\frac{\omega_0}{\omega_{\mathbf{q}}}(\cos\varphi,\ \sin\varphi,\ 0), \tag{246}$$

$$\boldsymbol{\epsilon}_{\mathrm{TM2}}=-i\,\frac{cq}{\omega_{2\mathbf{q}}}\,(0,0,1), \qquad \mathbf{d}=d\,(0,0,1), \tag{247}$$

$$\omega_{2\mathbf{q}}=\sqrt{4\omega_0^2+c^2q^2}=\sqrt{\omega_{\mathbf{q}}^2+3\omega_0^2}, \qquad \frac{c^2q^2}{\omega_{2\mathbf{q}}^2}=\frac{\omega_{\mathbf{q}}^2-\omega_0^2}{\omega_{\mathbf{q}}^2+3\omega_0^2}. \tag{248}$$

$$X_{\mathrm{TE1}}=-\sin\varphi\,h_x+\cos\varphi\,h_y, \qquad X_{\mathrm{TM1}}=\frac{\omega_0}{\omega_{\mathbf{q}}}\left(\cos\varphi\,h_x+\sin\varphi\,h_y\right), \qquad X_{\mathrm{TM2}}=0, \tag{249}$$

$$P_{\mathrm{TE1}}=0, \qquad P_{\mathrm{TM1}}=0, \qquad P_{\mathrm{TM2}}=-i\,\frac{cq}{\omega_{2\mathbf{q}}}\,d. \tag{250}$$

$$M^{\lambda 1}(\mathbf{k}',\mathbf{q})=\sqrt{\frac{A}{\omega_{\mathbf{q}}}}\,X_{\lambda 1}\begin{pmatrix}2t_a & 0\\ 0 & 2t_b\end{pmatrix}, \qquad \lambda=\mathrm{TE},\mathrm{TM}, \tag{251}$$

$$M^{\mathrm{TM2}}(\mathbf{k}',\mathbf{q})=\sqrt{\frac{A}{\omega_{2\mathbf{q}}}}\,\frac{cq}{\omega_{2\mathbf{q}}}\,d\begin{pmatrix}0 & \Delta_{ab}\\ \Delta_{ba} & 0\end{pmatrix}. \tag{252}$$

$$\widetilde{M}^{\lambda 1}_{\alpha\alpha}=\sqrt{\frac{A}{\omega_{\mathbf{q}}}}\,X_{\lambda 1}\,(2t_a uu'+2t_b vv'), \qquad \widetilde{M}^{\lambda 1}_{\alpha\beta}=\sqrt{\frac{A}{\omega_{\mathbf{q}}}}\,X_{\lambda 1}\,(2t_a uv'-2t_b vu'), \tag{253}$$

$$\widetilde{M}^{\lambda 1}_{\beta\alpha}=\sqrt{\frac{A}{\omega_{\mathbf{q}}}}\,X_{\lambda 1}\,(2t_a vu'-2t_b uv'), \qquad \widetilde{M}^{\lambda 1}_{\beta\beta}=\sqrt{\frac{A}{\omega_{\mathbf{q}}}}\,X_{\lambda 1}\,(2t_a vv'+2t_b uu'). \tag{254}$$

$$\widetilde{M}^{\mathrm{TM2}}_{\alpha\alpha}=-\sqrt{\frac{A}{\omega_{2\mathbf{q}}}}\,\frac{cq}{\omega_{2\mathbf{q}}}\,d\,(uv'\Delta_{ab}+vu'\Delta_{ba}), \qquad \widetilde{M}^{\mathrm{TM2}}_{\alpha\beta}=\sqrt{\frac{A}{\omega_{2\mathbf{q}}}}\,\frac{cq}{\omega_{2\mathbf{q}}}\,d\,(uu'\Delta_{ab}-vv'\Delta_{ba}), \tag{255}$$

$$\widetilde{M}^{\mathrm{TM2}}_{\beta\alpha}=\sqrt{\frac{A}{\omega_{2\mathbf{q}}}}\,\frac{cq}{\omega_{2\mathbf{q}}}\,d\,(uu'\Delta_{ba}-vv'\Delta_{ab}), \qquad \widetilde{M}^{\mathrm{TM2}}_{\beta\beta}=\sqrt{\frac{A}{\omega_{2\mathbf{q}}}}\,\frac{cq}{\omega_{2\mathbf{q}}}\,d\,(vu'\Delta_{ab}+uv'\Delta_{ba}). \tag{256}$$

$$\sum_{\lambda}\left|\widetilde{M}^{\lambda 1}_{\gamma\gamma'}\right|^2=\frac{A}{\omega_{\mathbf{q}}}\left(X^2_{\mathrm{TE1}}+X^2_{\mathrm{TM1}}\right)C^2_{\gamma\gamma'},\qquad \left|\widetilde{M}^{\mathrm{TM2}}_{\gamma\gamma'}\right|^2=\frac{A}{\omega_{2\mathbf{q}}}\,\frac{\omega_{\mathbf{q}}^2-\omega_0^2}{\omega_{\mathbf{q}}^2+3\omega_0^2}\,D^2_{\gamma\gamma'}. \tag{257}$$

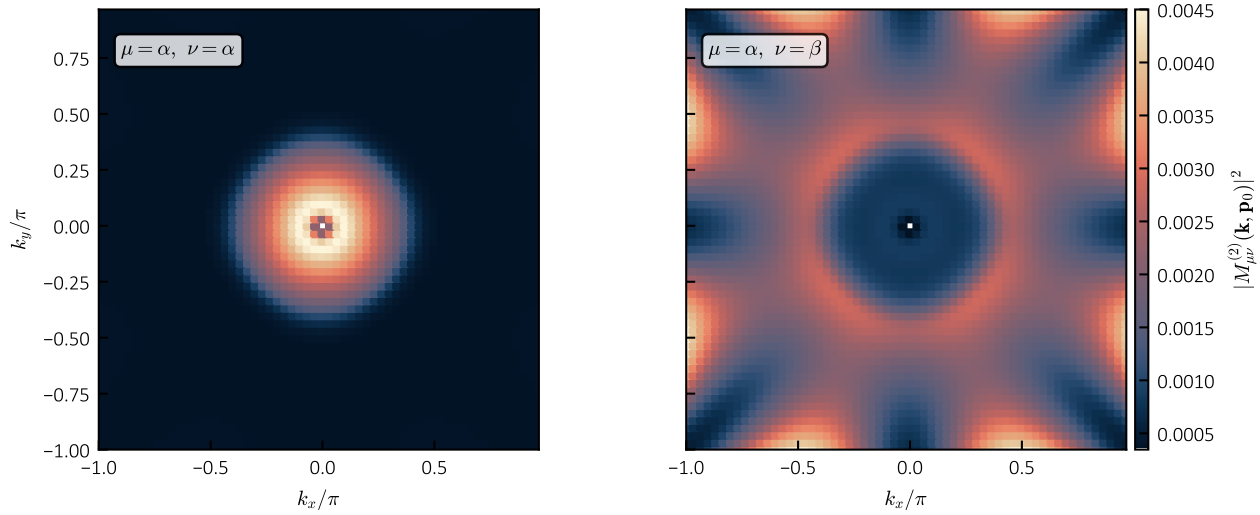
$$\begin{aligned}\Gamma_{\gamma'\mathbf{k}'\to\gamma\mathbf{k}}=A\,\Omega N(\Omega)\Bigg[\Bigg(\left(\sin^2\varphi+\frac{\omega_0^2}{\omega_{\mathbf{q}}^2}\cos^2\varphi\right)h_x^2+\left(\cos^2\varphi+\frac{\omega_0^2}{\omega_{\mathbf{q}}^2}\sin^2\varphi\right)h_y^2+\\+2\left(\frac{\omega_0^2}{\omega_{\mathbf{q}}^2}-1\right)\cos\varphi\sin\varphi\,h_xh_y\Bigg)C^2_{\gamma\gamma'}\,\frac{8\gamma}{\left[(\Omega-\omega_{\mathbf{q}})^2+\gamma^2\right]\left[(\Omega+\omega_{\mathbf{q}})^2+\gamma^2\right]}+\\+\frac{\omega_{\mathbf{q}}^2-\omega_0^2}{\omega_{\mathbf{q}}^2+3\omega_0^2}\,D^2_{\gamma\gamma'}\,\frac{8\gamma}{\left[(\Omega-\omega_{2\mathbf{q}})^2+\gamma^2\right]\left[(\Omega+\omega_{2\mathbf{q}})^2+\gamma^2\right]}\Bigg].\end{aligned} \tag{258}$$

Minimal coupling v limiti $q\to 0$:

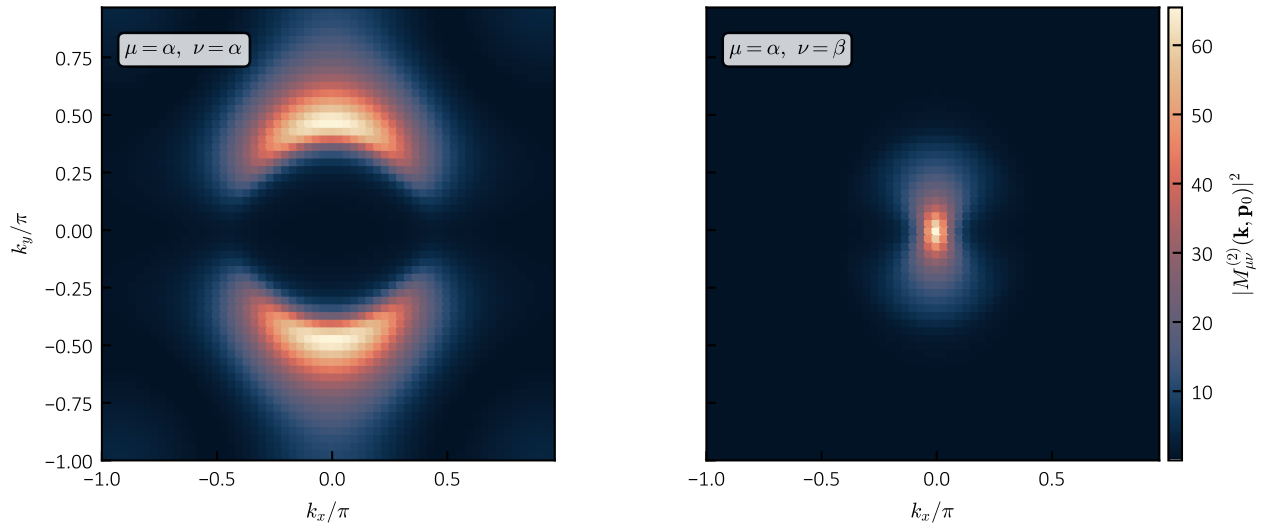
$$\Gamma_{\gamma'\mathbf{k}'\to\gamma\mathbf{k}}=A\left[\sin^2\left(k'_x+\tfrac{q_x}{2}\right)+\sin^2\left(k'_y+\tfrac{q_y}{2}\right)\right]C^2_{\gamma\gamma'}\cdot\Omega N(\Omega)\cdot\frac{8\gamma}{\left[(\Omega-\omega_{\mathbf{q}})^2+\gamma^2\right]\left[(\Omega+\omega_{\mathbf{q}})^2+\gamma^2\right]}. \tag{259}$$

4 Vizualizacije kopeli za fotone

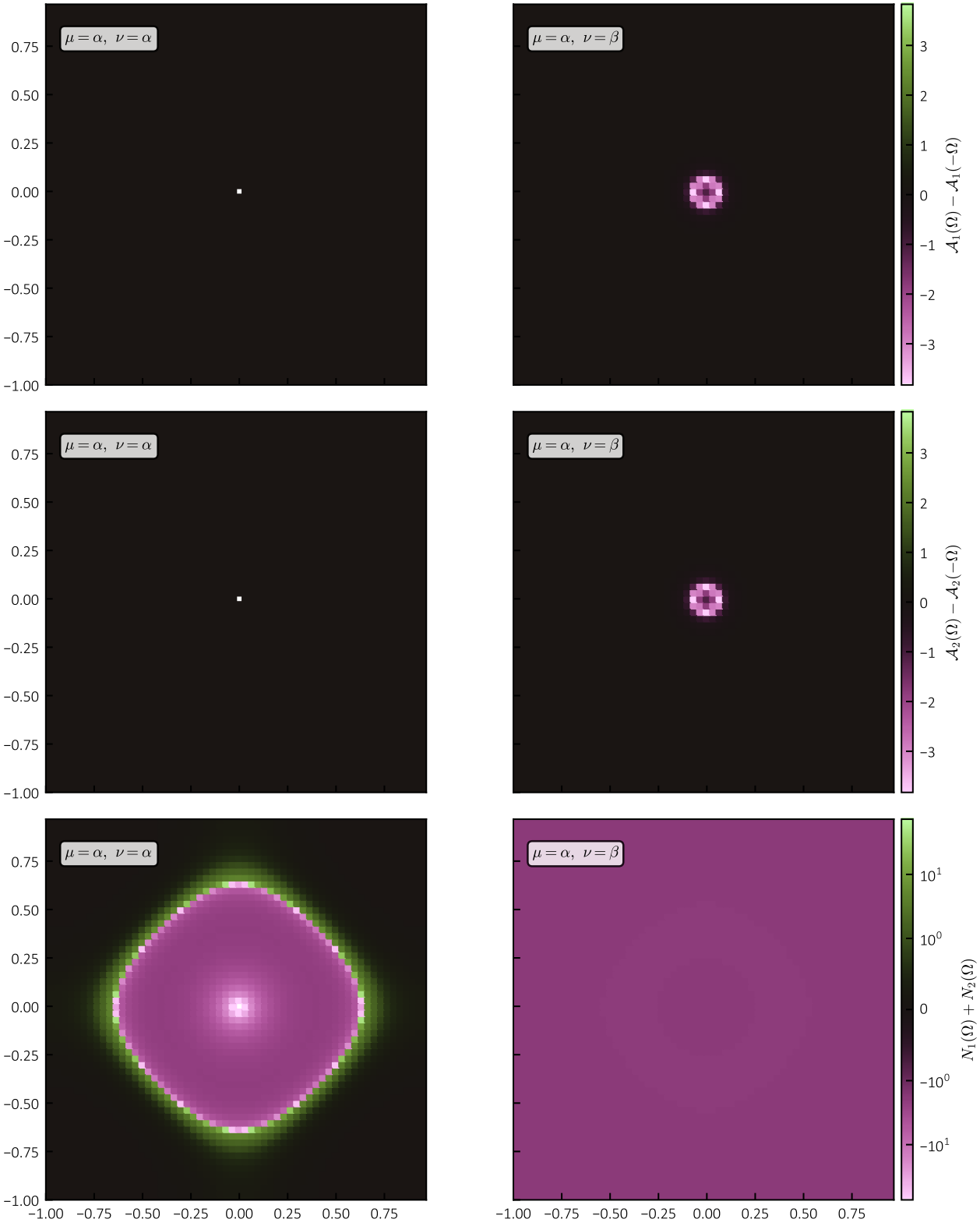
Vsi naslednji grafi so v tocki $T_1=1.2T_c, T_2=0.2T_c$ in uporabljajo fotonski verteks, kot opisano zgoraj, $\mathbf{q}=\mathbf{k}-\mathbf{p}_0$.



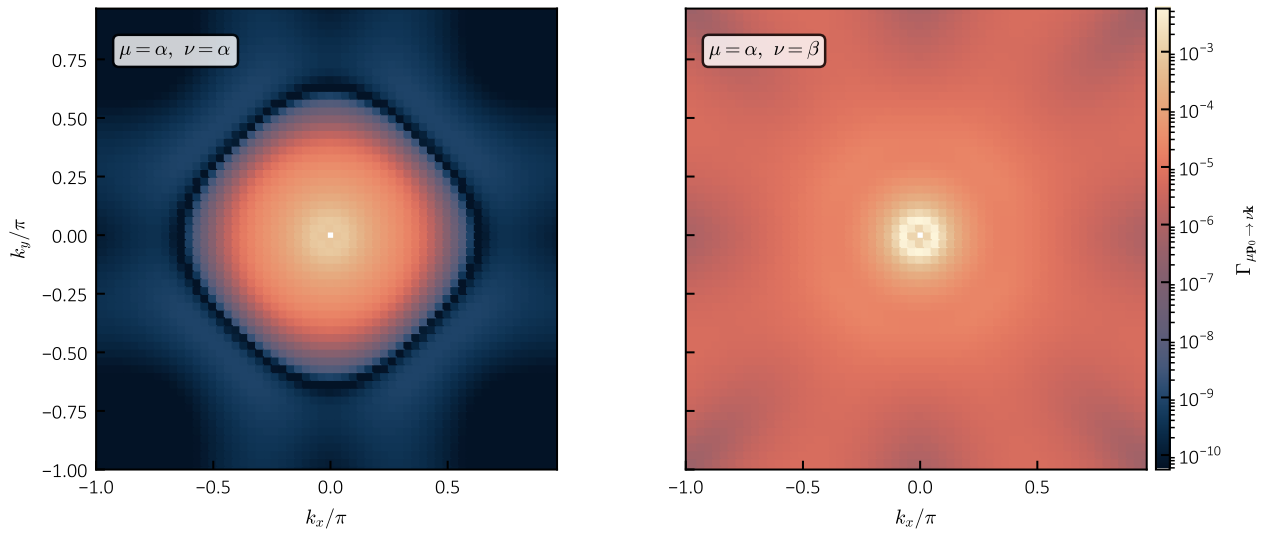
Slika 2 | Verteks pri $\mathbf{d}=0$ za $\mathbf{p}_0=(0,0)$ za obe kopeli z disperzijo: $\omega_{\mathbf{q}}=\sqrt{\omega_0^2+c^2q^2}$, $c=10, \omega_0=2.2$, indeks gor je indeks kopeli (za obe je tu enaka disperzija).



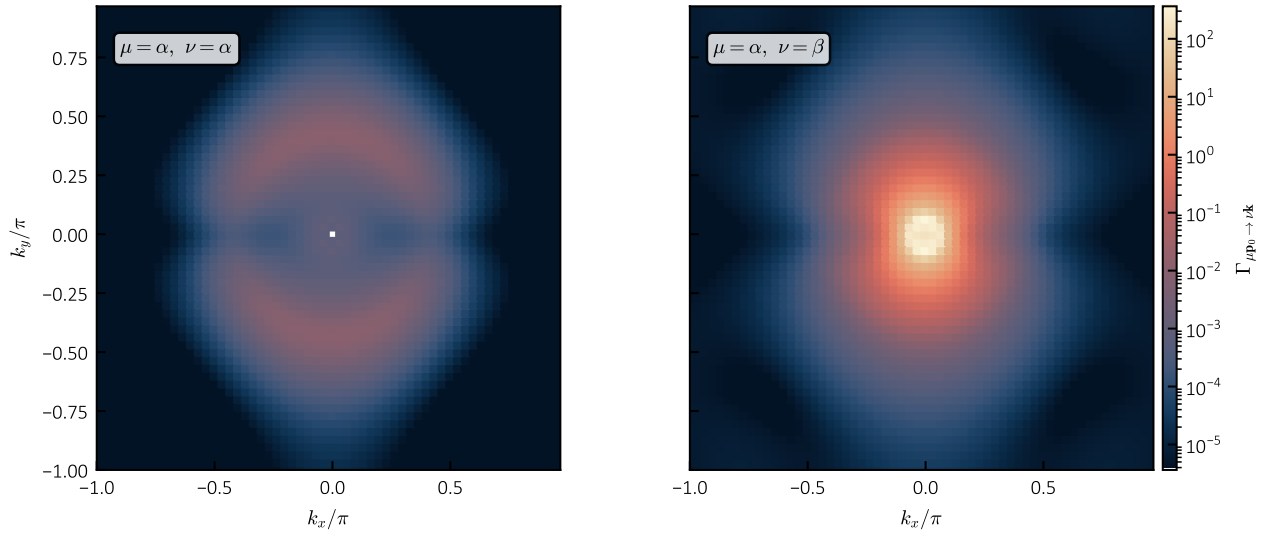
Slika 3 | Verteks pri $\mathbf{d} = (2, 0, 0)$ v x smeri za $\mathbf{p}_0 = (0,0)$ za obe kopeli z disperzijo: $\omega_{\mathbf{q}} = \sqrt{\omega_0^2 + c^2 q^2}$, $c = 10, \omega_0 = 2.2$. Dipol zlomi $C_{4v} \to C_{2v}$ simetrijo.



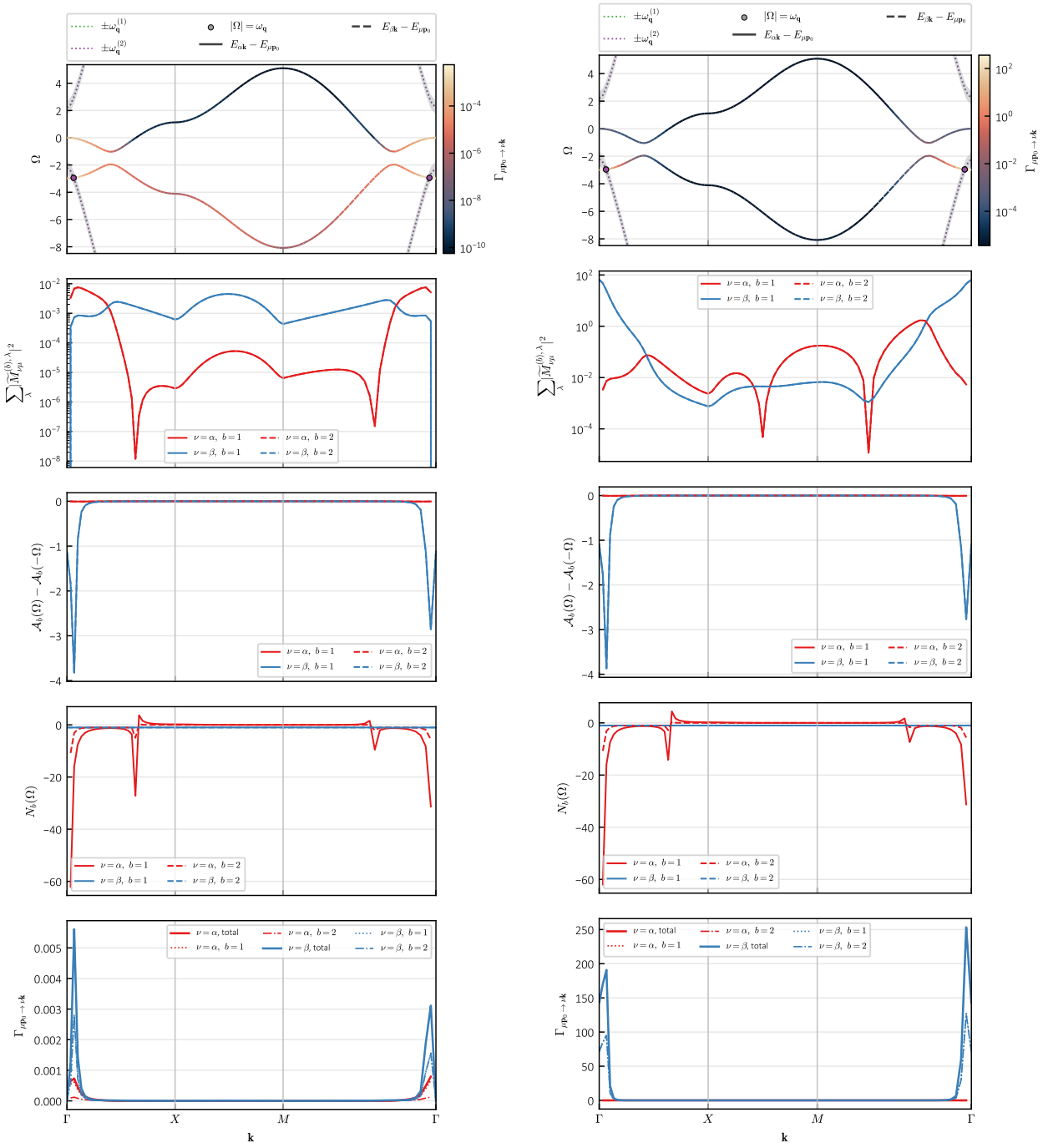
Slika 4 | Komponente faktorjev za kopeli pri $\omega_{\mathbf{q}} = \sqrt{\omega_0^2 + c^2 q^2}$, $c = 10, \omega_0 = 2.2$ (majhni q prehodi).



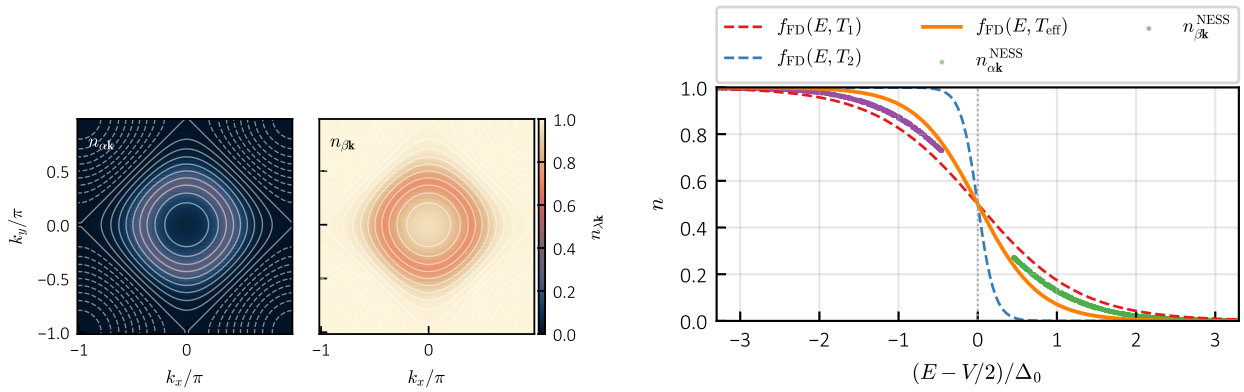
Slika 5 | Končni transfer rate pri $\omega_{\mathbf{q}} = \sqrt{\omega_0^2 + c^2 q^2}$, $c = 10, \omega_0 = 2.2$, $\mathbf{d} = (0, 0, 0)$.



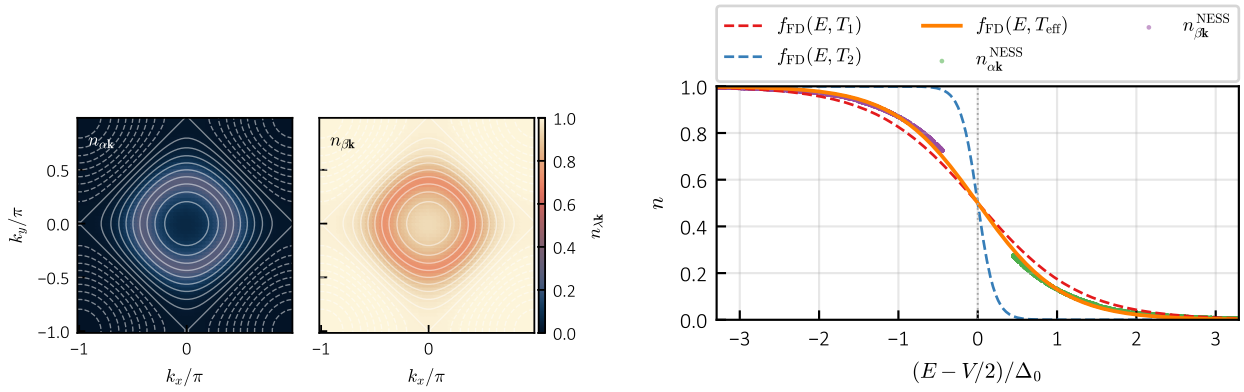
Slika 6 | Končni transfer rate pri $\omega_{\mathbf{q}} = \sqrt{\omega_0^2 + c^2 q^2}$, $c = 10, \omega_0 = 2.2$, $\mathbf{d} = 2(1, 0, 0)$.



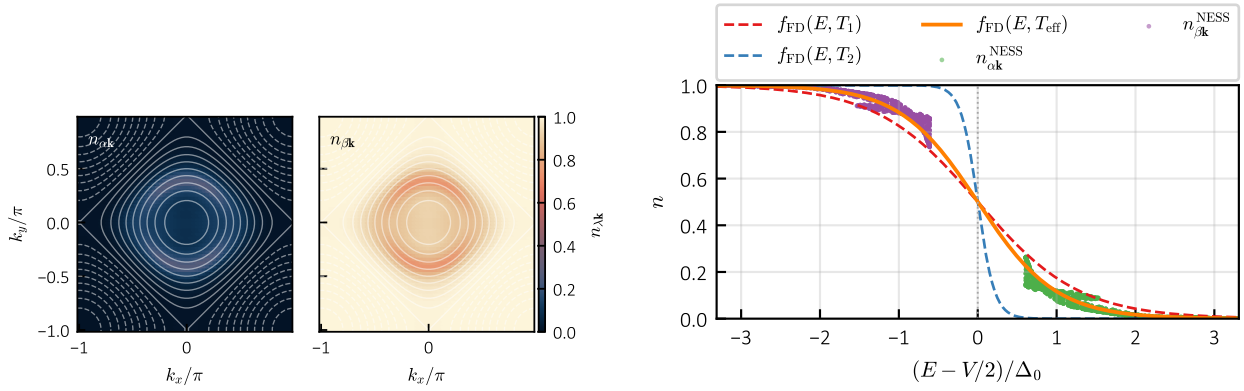
Slika 7 | Komponente faktorjev za kopeli pri $\omega_{\mathbf{q}} = \sqrt{\omega_0^2 + c^2 q^2}$, $c = 10, \omega_0 = 2.2$ (majhni q prehodi).



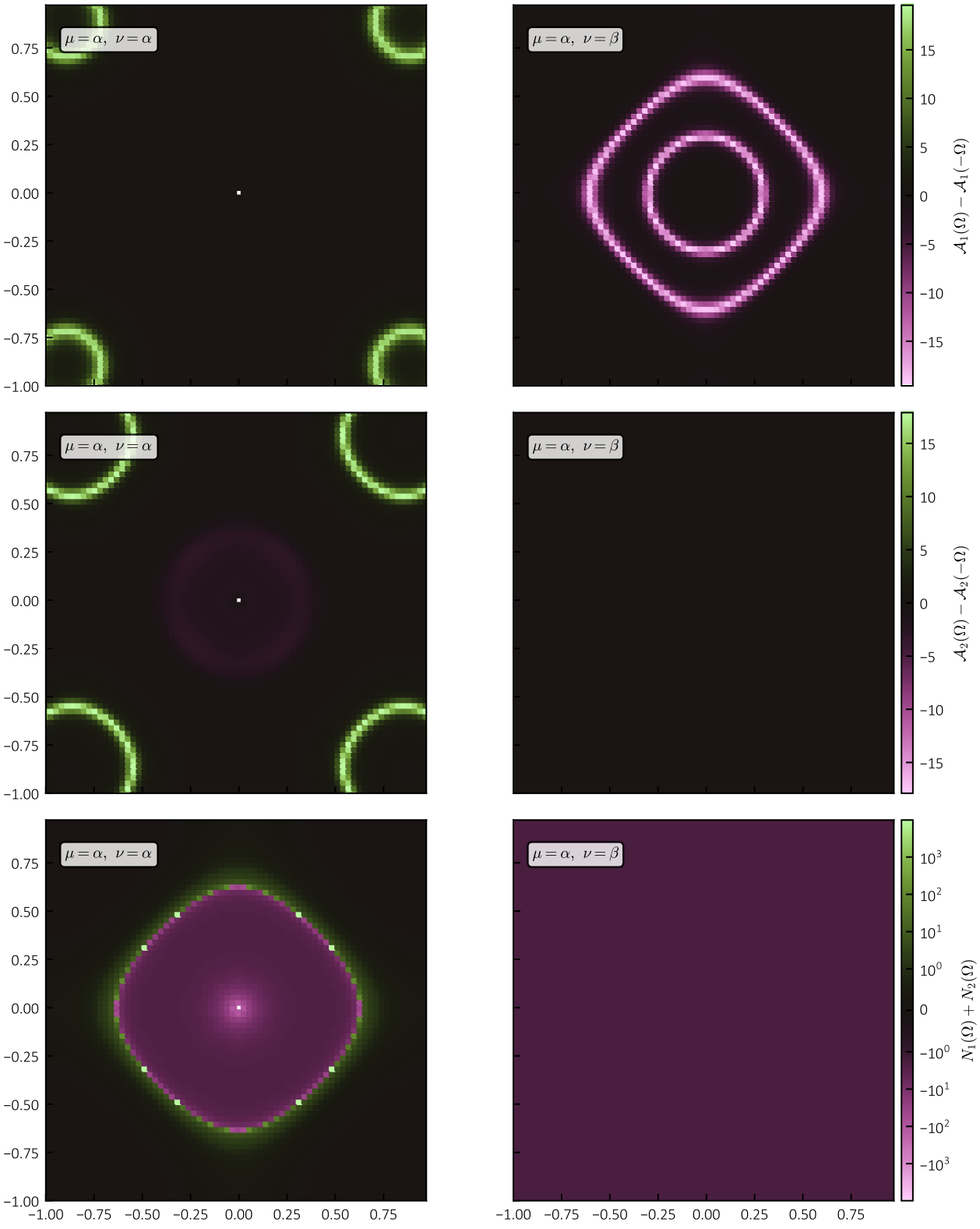
Slika 8 | Komponente faktorjev za kopeli pri $\omega_{\mathbf{q}} = \sqrt{\omega_0^2 + c^2 q^2}$, $c = 10, \omega_0 = 2.2$ (majhni q prehodi).



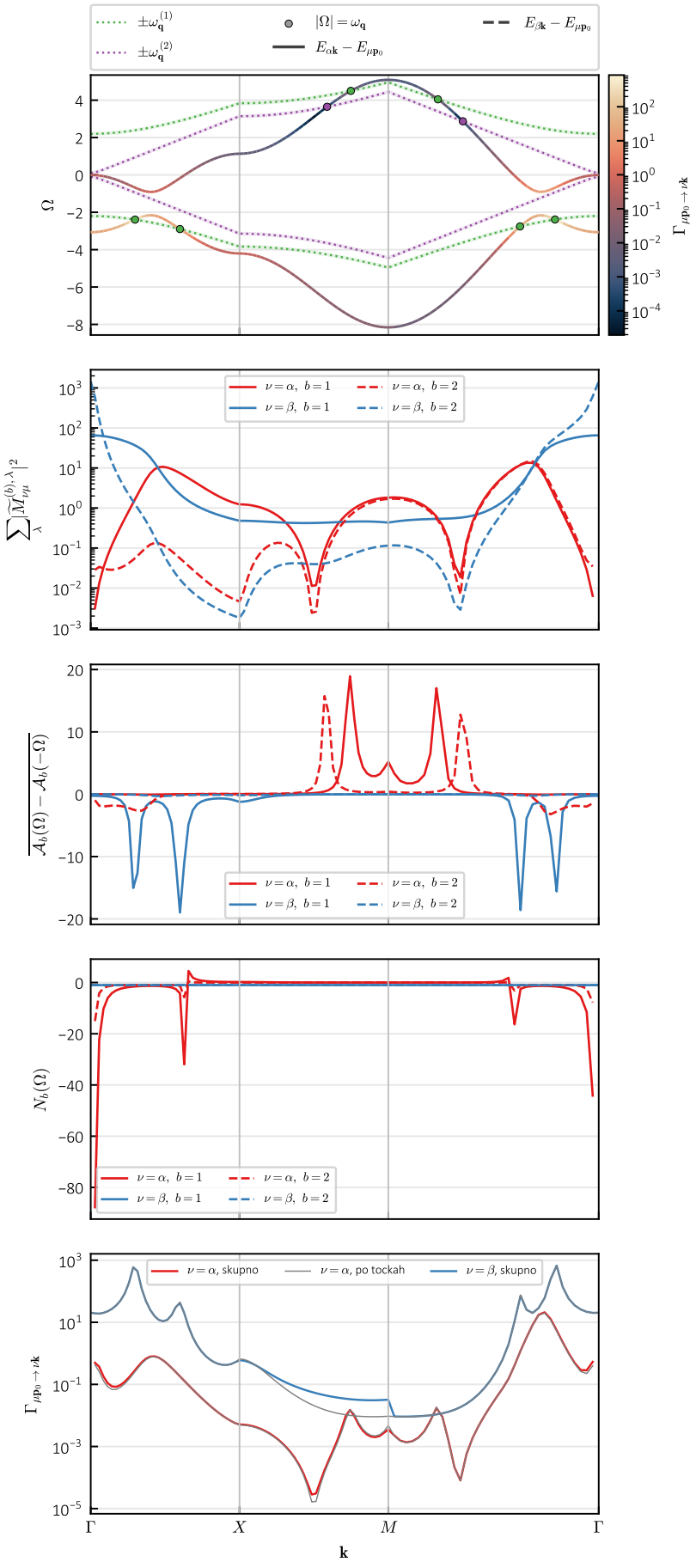
Slika 9 | Komponente faktorjev za kopeli pri $\omega_{\mathbf{q}} = \sqrt{\omega_0^2 + c^2 q^2}$, $c = 10, \omega_0 = 2.2$ (majhni q prehodi).



Slika 10 | Komponente faktorjev za kopeli pri mixed disperzijah: $c = 1.0, \omega_0^{(2)} = 2.2, \omega_0^{(1)} = 0.1$ in $\mathbf{d} = 2(1,0,0)$ zdaj zlomi simetrijo v $n(s) \to n(s,u)$, zgleda kar C_{2v}.



Slika 11 | Komponente faktorjev za kopeli pri $c = 1.0, \omega_0^{(1)} = 2.2, \omega_0^{(2)} = 0.1$ in $\mathbf{d} = 2(1, 0, 0)$.



Slika 12 | Mixed disperzije: $c = 1.0, \omega_0^{(1)} = 2.2, \omega_0^{(2)} = 0.1$ in $\mathbf{d} = 2(1, 0, 0)$.

4.1 Numerika detajli (konvergenca in povprecenje)

Za manjše mreže nas zanima kaj se zgodi, če zgrešimo vrh resonance, ker je energija poračunana v neenakomerno razporejenih točkah. Zaradi tega lahko pogledamo, kaj se zgodi s povprečenjem čez sosednje točke. V 1D primeru je mreža definirana v točkah

$$k_i = i\,\delta k, \qquad \mathcal{C}_i = \left[k_i - \tfrac{\delta k}{2}, k_i + \tfrac{\delta k}{2}\right], \qquad q_{ij} = k_i - k_j, \qquad \Omega_{ij} = E_\nu(k_i) - E_\mu(k_j), \tag{260}$$

$$\mathcal{S}(\Omega,\omega) = \mathcal{A}(\Omega) - A(-\Omega) = \frac{8\gamma\,\omega\,\Omega}{\left[(\Omega-\omega)^2+\gamma^2\right]\left[(\Omega+\omega)^2+\gamma^2\right]}\,, \qquad \omega_q = \sqrt{\omega_0^2 + c_s^2 q^2}\,. \tag{261}$$

Po točkah bi računali:

$$\mathcal{S}_{ij}^{\mathrm{pt}} = \mathcal{S}\big(\Omega_{ij}, \omega_{q_{ij}}\big). \tag{262}$$

V svojih celicah sta k in p porazdeljena enakomerno, zato lahko napišemo povprečje čez celico kot

$$\overline{\mathcal{S}}_{ij} = \frac{1}{\delta k^2}\int_{\mathcal{C}_i}\mathrm{d}k\int_{\mathcal{C}_j}\mathrm{d}p\;\mathcal{S}\big(E_\nu(k) - E_\mu(p), \omega_{k-p}\big) = \int\mathrm{d}\Omega\int\mathrm{d}q\;\rho_{ij}(\Omega,q)\,\mathcal{S}(\Omega,\omega_q)\,, \tag{263}$$

z vrjetnostno gostoto:

$$\rho_{ij}(\Omega,q) = \frac{1}{\delta k^2}\int_{\mathcal{C}_i}\mathrm{d}k\int_{\mathcal{C}_j}\mathrm{d}p\;\delta\big(\Omega - E_\nu(k) + E_\mu(p)\big)\,\delta\big(q - k + p\big). \tag{264}$$

Naredil sem približek, kjer razlike energij odčitamo točno v diskretnih točkah, povprečimo pa čez momente (skupna verjentostna gostota za razliko dveh enakomernih spremeljivk $q = k - p$ je konvolucija, torej trikotna funkcija P)

$$\rho_{ij}(\Omega,q) \;\approx\; \delta(\Omega - \Omega_{ij})\,P(q - q_{ij}), \qquad P(x) = \frac{1}{\delta k}\Big(1 - \frac{|x|}{\delta k}\Big)\,\Theta(\delta k - |x|), \qquad \int P = 1\,, \tag{265}$$

kar nam da skupaj pri fiksni energijski razliki $\Omega_{ij} = E_\nu(k_i) - E_\mu(p_j)$:

$$\overline{\mathcal{S}}_{ij} \approx \int_{q_{ij}-\delta k}^{q_{ij}+\delta k} dq\;P(q - q_{ij})\,\mathcal{S}(\Omega_{ij},\omega_q) \tag{266}$$

V praksi to pomeni, da vzamemo n_s točk in seštejemo namest integriramo:

$$t_l = 2\Big[\frac{l-\frac{1}{2}}{n_s} - \frac{1}{2}\Big], \qquad , \qquad \sum_{l=1}^{n_s}\frac{2}{n_s}\big(1 - |t_l|\big) = 1\,(n_s\ \mathrm{sod}), \qquad l = 1,\ldots,n_s, \tag{267}$$

oz. diskretno (z upeljavo $t = (q - q_{ij})/\delta k$) za arbitrarno funkcijo $F(q)$:

$$\int_{-1}^{1}\mathrm{d}t\;(1-|t|)\,F(q_{ij} + t\delta k) = \sum_l \frac{2}{n_s}\big(1 - |t_l|\big)F(q_{ij} + t_l\delta k) + \mathcal{O}(n_s^{-2}). \tag{268}$$

V principu se lahko tu ustavimo, ampak je računanje za vsak par Ω_{ij} zahtevno $\mathcal{O}((2N_k^2)^2 n_s^2)$ (veliko parov $\mathbf{q}_{ij}$ da enako energijo, ta pa se z iteracijami spreminja, zato ga tabeliram in v vsaki iteraciji odčitam). Zato tabeliramo razlike energij in interpoliramo. Izbral sem $\delta\Omega = \gamma/8$, tako je $N_\Omega = \lceil E_{\max}/\delta\Omega\rceil + 2$, kjer si $E_{\max}$ izberemo tako, da je večji od največje možne razlike energij[1]. Za neko funkcijo $F(\Omega)$ potem računamo le N_Ω točk in za določen $\Omega_n \le \Omega_{ij} \le \Omega_{n+1}$ interpoliramo linearno:

$$F(\Omega) = (1-\alpha)F(\Omega_n) + \alpha F(\Omega_{n+1})\,, \quad \alpha = |\Omega|/\delta\Omega - \lfloor|\Omega|/\delta\Omega\rfloor \tag{269}$$

to skupaj stane $N_k^2 N_\Omega n_s^2 + (2N_k)^2$, dodatni člen je za odčitavanje energije. Na koncu se bolj splača skupaj povprečiti ne le spektralno funkcijo, temveč tudi dodatne faktorje. S krajšo notacijo:

$$\mathbf{h} = \Big(\sin\big(k'_x + \tfrac{q_x}{2}\big),\ \sin\big(k'_y + \tfrac{q_y}{2}\big)\Big) \tag{270}$$

in

$$Z_{xx} = \sin^2\varphi + (\omega_0/\omega_{\mathbf{q}})^2\cos^2\varphi, \qquad Z_{yy} = \cos^2\varphi + (\omega_0/\omega_{\mathbf{q}})^2\sin^2\varphi, \qquad Z_{xy} = 2\sin\varphi\cos\varphi\,((\omega_0/\omega_{\mathbf{q}})^2 - 1), \tag{271}$$

[1] V kodi to ocenim avtomatsko navzgor z 10% maržo.

definiramo še (bolj stabilno v kodi, tudi produkt $\Omega N(\Omega)$ je stabilen za $\Omega \to 0$):

$$F_{\mathbf{q}}(\Omega) = \frac{\mathcal{A}_{\mathbf{q}}(\Omega) - \mathcal{A}_{\mathbf{q}}(-\Omega)}{\Omega\omega_{\mathbf{q}}} = \frac{8\gamma}{\left[(\Omega-\omega)^2+\gamma^2\right]\left[(\Omega+\omega)^2+\gamma^2\right]}\,, \tag{272}$$

tako je po točkah (to je enaka matrika kot (245))

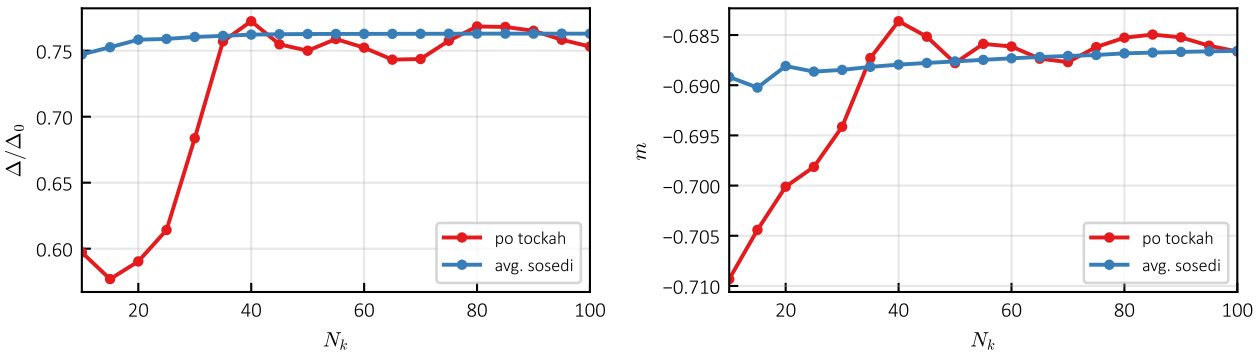
$$\Gamma_{\gamma'\mathbf{k}'\to\gamma\mathbf{k}} = A\cdot\Omega N(\Omega)\cdot\left[\left(h_x^2 Z_{xx}F_{\mathbf{q}} + h_y^2 Z_{yy}F_{\mathbf{q}} + h_x h_y Z_{xy}F_{\mathbf{q}}\right)C^2_{\gamma\gamma'} + Z_{xx}F_{\mathbf{q}}D^2_{\gamma\gamma'}\right], \tag{273}$$

Povprečimo produkte tipa $Z_{xx}F_{\mathbf{q}}$ s trikotno funkcijo, medtem ko $\mathbf{h}$, $C_{\gamma,\gamma'}$ in Ω računamo le v centrih celic. Povprečenje $N(\Omega)$ po energiji bi lahko zlomilo detailed balance glede na mrežo. Faktorji $\mathbf{h}$ v argumentu ne vsebujejo razlike momentov (če je $\mathbf{k} = \mathbf{k}_i + \mathbf{x}$ in $\mathbf{k}' = \mathbf{k}'_j + \mathbf{y}$ kjer sta $\mathbf{x}$ in $\mathbf{y}$ naključni spremenljivki v celici, potem povprečimo po $\mathbf{x}-\mathbf{y}$, argument $\mathbf{h}$ pa je $(\mathbf{k}_i + \mathbf{k}'_j)/2 + (\mathbf{x}+\mathbf{y})/2$)

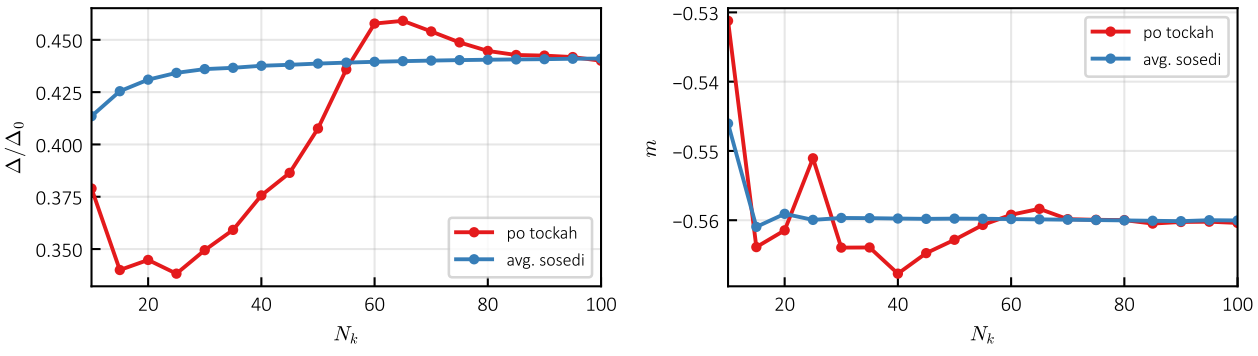
$$\overline{\Gamma}_{\gamma'\mathbf{k}'\to\gamma\mathbf{k}} = A\cdot\Omega N(\Omega)\cdot\left[\left(h_x^2\overline{Z_{xx}F_{\mathbf{q}}} + h_y^2\overline{Z_{yy}F_{\mathbf{q}}} + h_x h_y\overline{Z_{xy}F_{\mathbf{q}}}\right)C^2_{\gamma\gamma'} + \overline{Z_{xx}F_{\mathbf{q}}}D^2_{\gamma\gamma'}\right], \tag{274}$$

kjer je

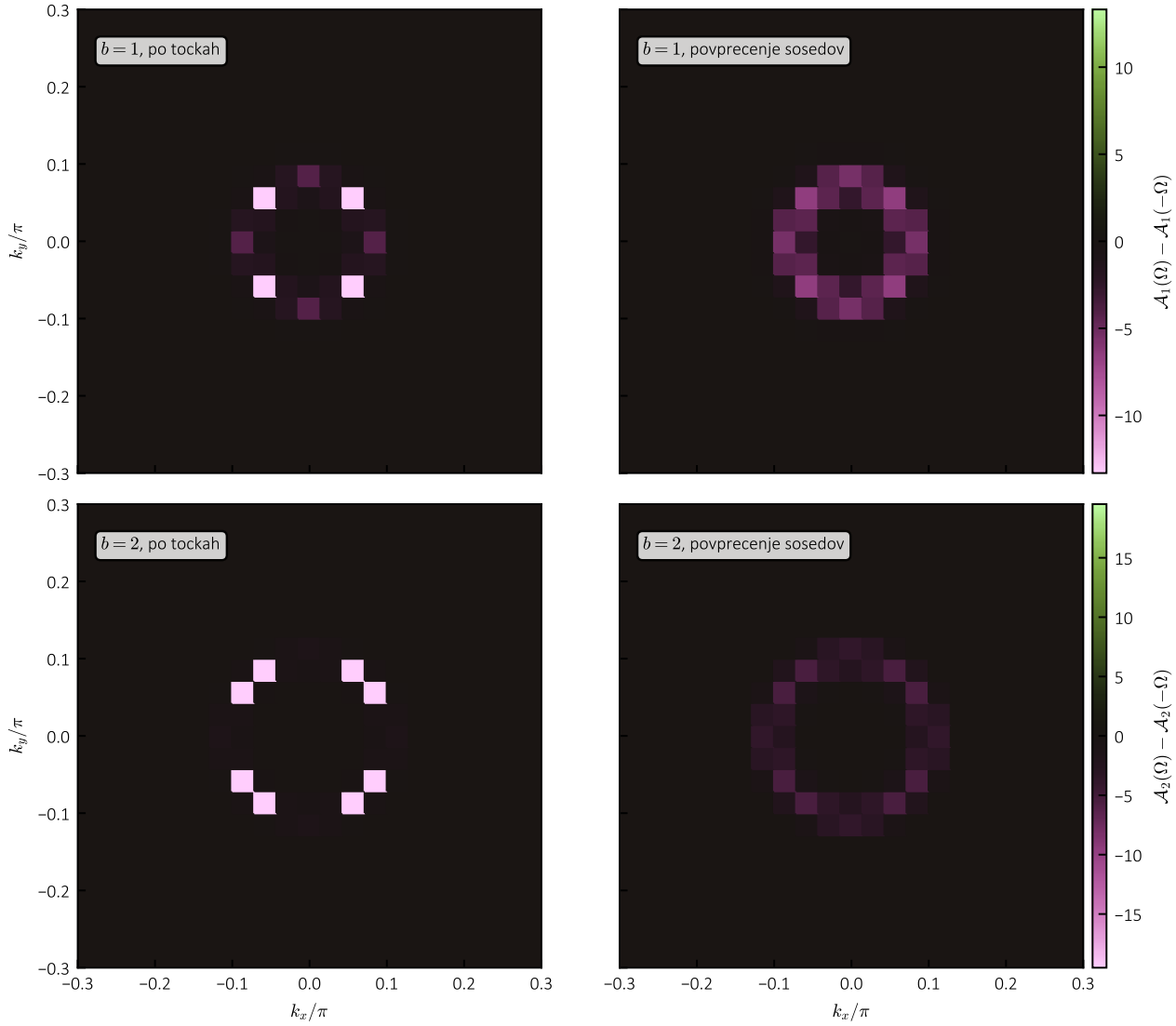
$$\overline{X_{\mathbf{q}}(\Omega)} = \int \mathrm{d}x\,\mathrm{d}y\,P(x)P(y)\,X_{(q_x+x,q_y+y)} \simeq \sum_{l,m}^{n_s}\left(\frac{2}{n_s}\right)^2\left(1-|t_l|\right)\left(1-|t_m|\right)X_{\mathbf{q}+(t_l\delta k_x,\,t_m\delta k_y)}\,. \tag{275}$$



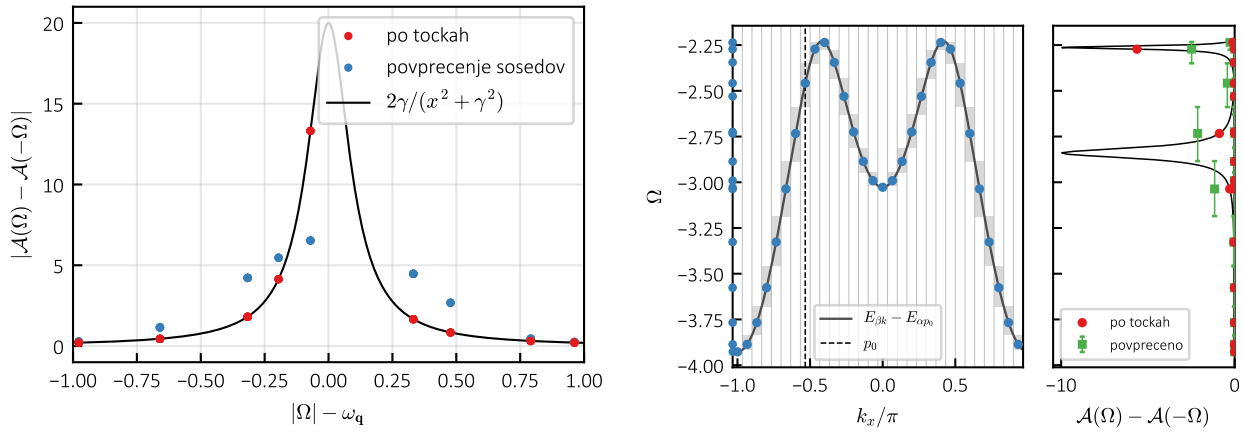
Slika 13 | Mixed disperzije: $c = 10.0$, $\omega_0^{(1)} = 2.2$, $\omega_0^{(2)} = 0.1$ in $\mathbf{d} = 2(1,0,0)$ ter $\gamma = 0.1$. KOnvergenca pri $T_1 = 1.2T_c$ in $T_2 = 0.2T_c$. TO je za primer ko je $q \ll k_F$, tako da so small momentum prehodi, vecinoma interband (diagonalna baza).



Slika 14 | Mixed disperzije: $c = 10.0$, $\omega_0^{(1)} = 2.2$, $\omega_0^{(2)} = 0.1$ in $\mathbf{d} = 2(1,0,0)$ ter $\gamma = 0.1$. KOnvergenca pri $T_1 = 2.5T_c$ in $T_2 = 0.2T_c$. TO je za primer ko je $q \ll k_F$, tako da so small momentum prehodi, vecinoma interband (diagonalna baza) (dlje od diagonale $T_1 = T_2$ pocasneje konvergira).



Slika 15 | .



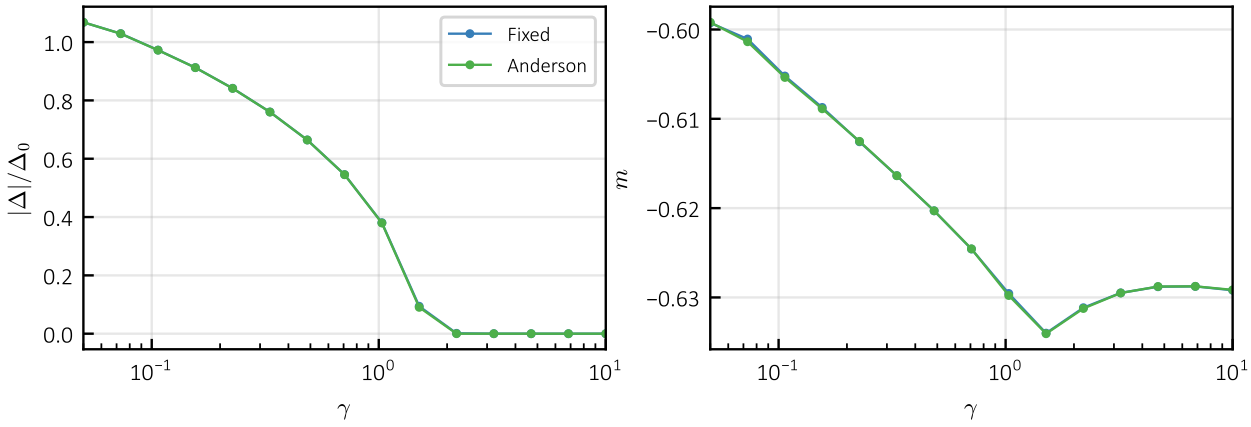
Slika 16 | .

4.1.1 Andersonov algoritem

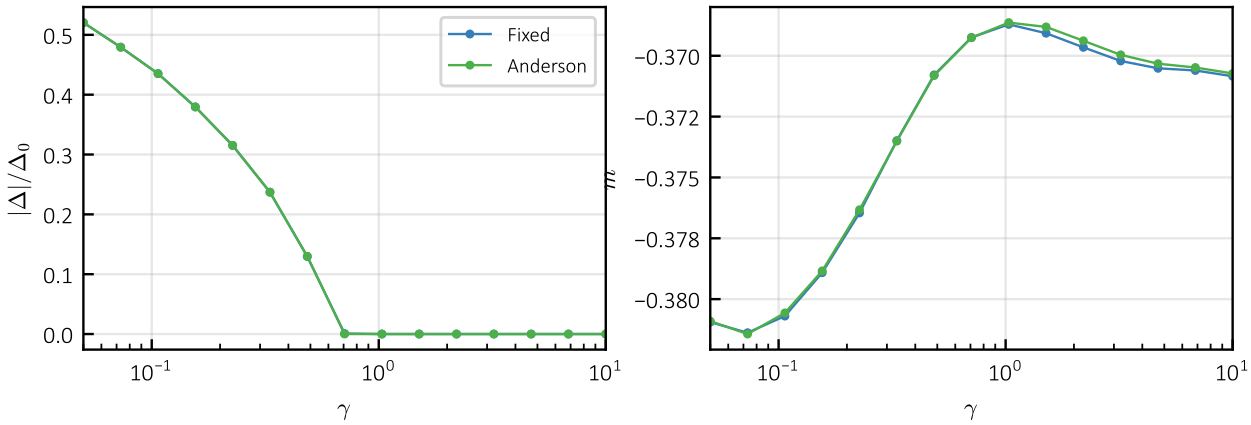
Samokonsistenten sistem enačb je ekvivalenten reševanju fixed point problema $\mathbf{x} = \mathcal{G}(\mathbf{x})$, oziroma iskanje ničel funckije $\mathbf{f}(\mathbf{x}) = \mathcal{G}(\mathbf{x}) - \mathbf{x}$. Pri Picard iteracijah samo posodabljamo s prejšnjo vrednostjo:

$$\mathbf{x}_{k+1} = \mathbf{x}_k + \beta\, \mathbf{f}_k \,. \tag{276}$$

okoli rešitve se konvergenca ustavi, saj je: $\mathbf{f}(\mathbf{x}) \approx J(\mathbf{x} - \mathbf{x}^*)$ kjer je Jakobijan $J = \partial\mathcal{G}/\partial\mathbf{x} - \mathbb{1}$. ... (see section 3.)



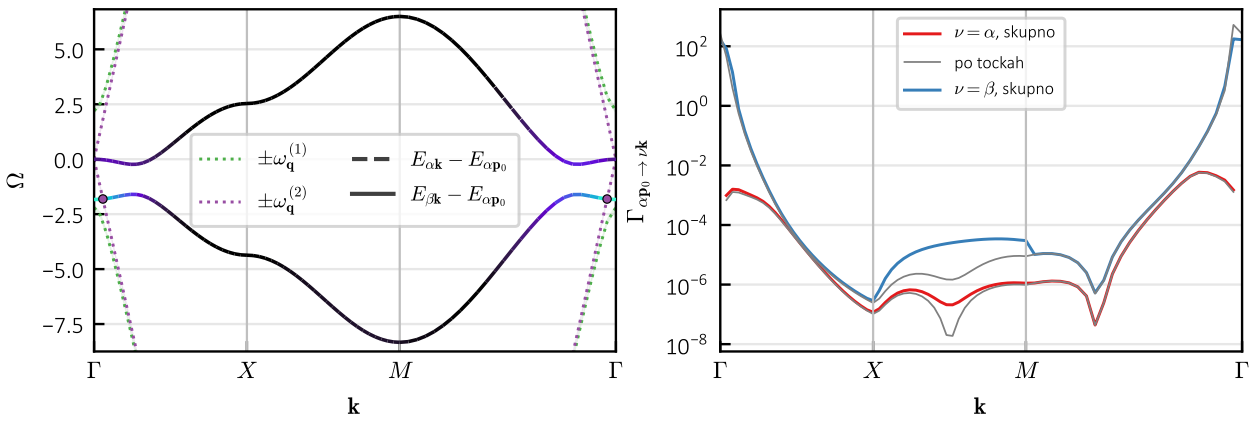
Slika 17 | $G = 2, t_a = -t_b = 1$ in $V = 7.8$, kjer je $\Delta_0 = 1.0$. Disperziji sta: $c = 10.0$, $\omega_0^{(1)} = 3.0$, $\omega_0^{(2)} = 0.01$. V sklopitvi je $d = 2$ in $N_k = 65$, $n_s = 8$. V točki $T_1 = 4.4T_c$ in $T_2 = 0.01T_c$.



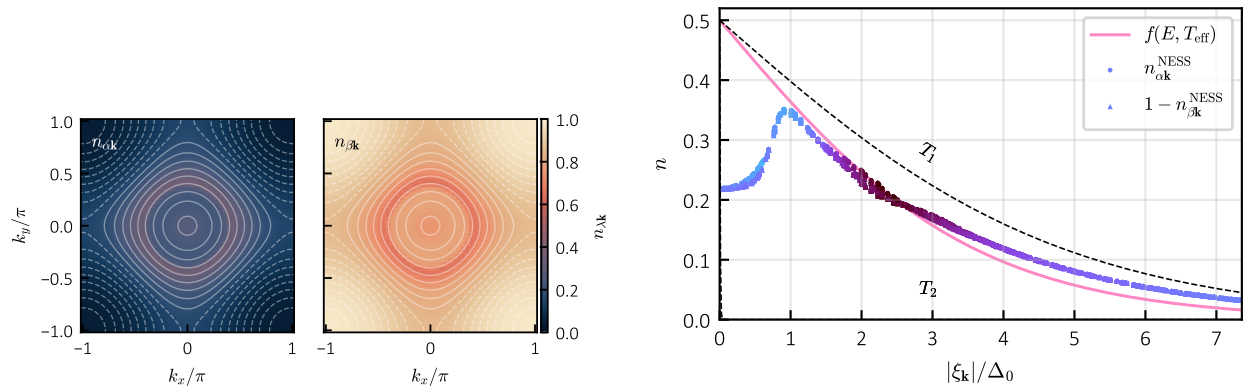
Slika 18 | $G = 2, t_a = -t_b = 1$ in $V = 4.4$, kjer je $\Delta_0 = 1.0$. Disperziji sta: $c = 10.0$, $\omega_0^{(1)} = 3.0$, $\omega_0^{(2)} = 0.01$. V sklopitvi je $d = 2$ in $N_k = 65$, $n_s = 8$. V točki $T_1 = 4.4T_c$ in $T_2 = 0.01T_c$.

4.2 Primer NESS za BEC

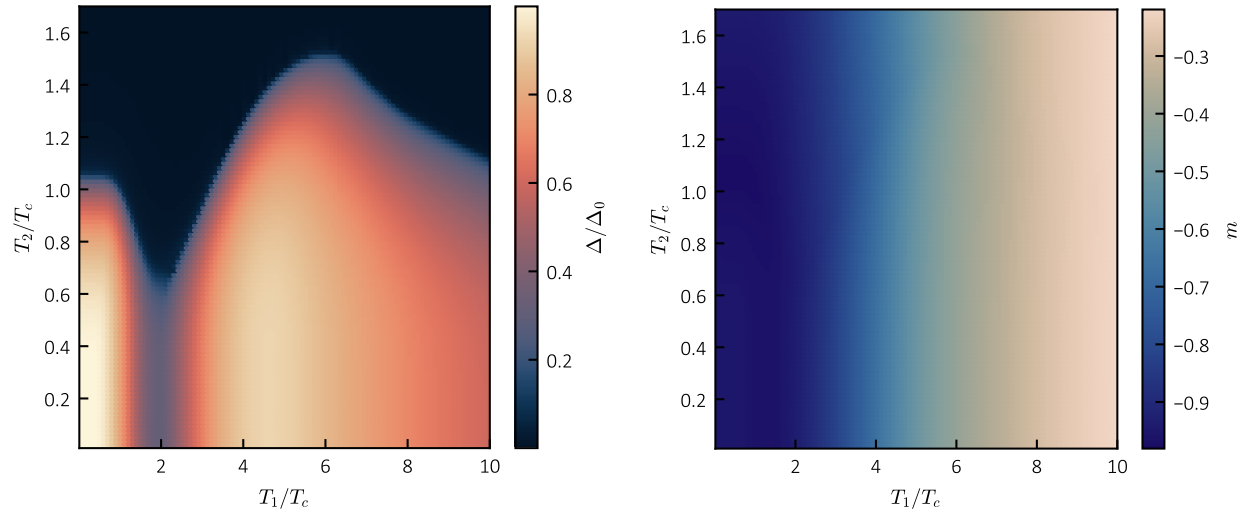
Primer NESS v BEC režimu: $G = 2, t_a = -t_b = 1$ in $V = 7.8$, kjer je $\Delta_0 = 1.0$. Disperziji sta: $c = 10.0$, $\omega_0^{(1)} = 2.2$, $\omega_0^{(2)} = 0.01$. V sklopitvi je $d = 2$ in $\gamma = 0.15$ in $N_k = 50$.



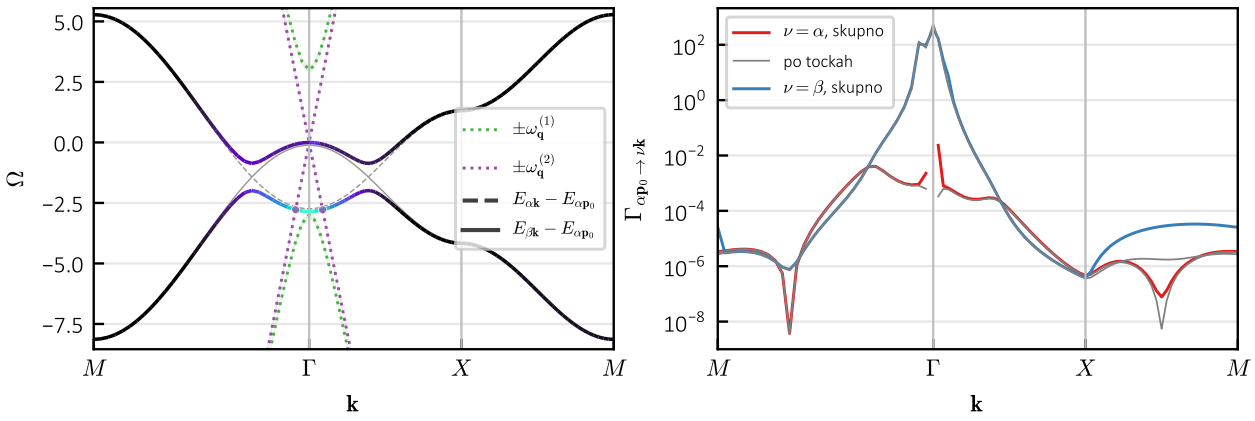
Slika 19 | Grafi so v točki $T_1 = 4.4T_c$ in $T_2 = 0.01T_c$, kjer preidemo v BCS režim.



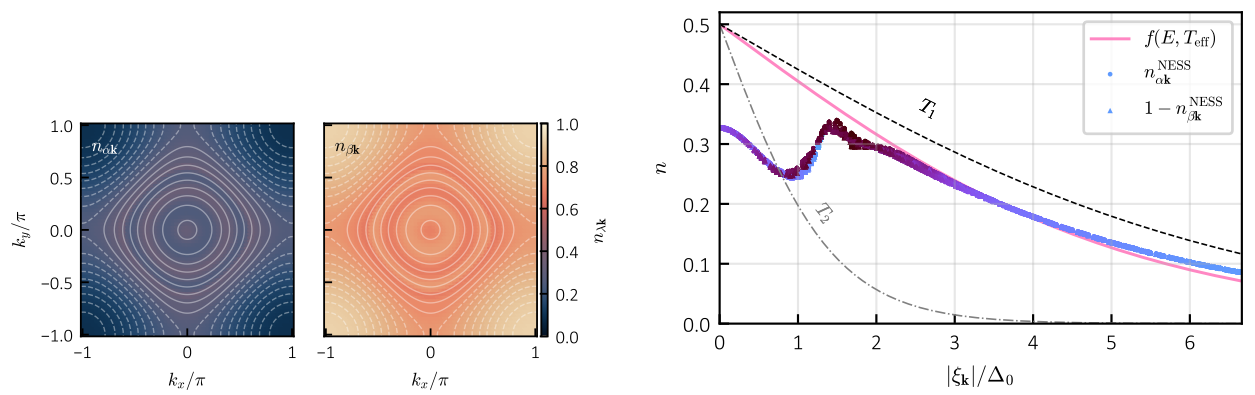
Slika 20 | Grafi so v točki $T_1 = 4.4T_c$ in $T_2 = 0.01T_c$, kjer preidemo v BCS režim.



Slika 21 | Disperziji sta: $c = 10.0, \omega_0^{(1)} = 3, \omega_0^{(2)} = 0.01$, BEC režim.



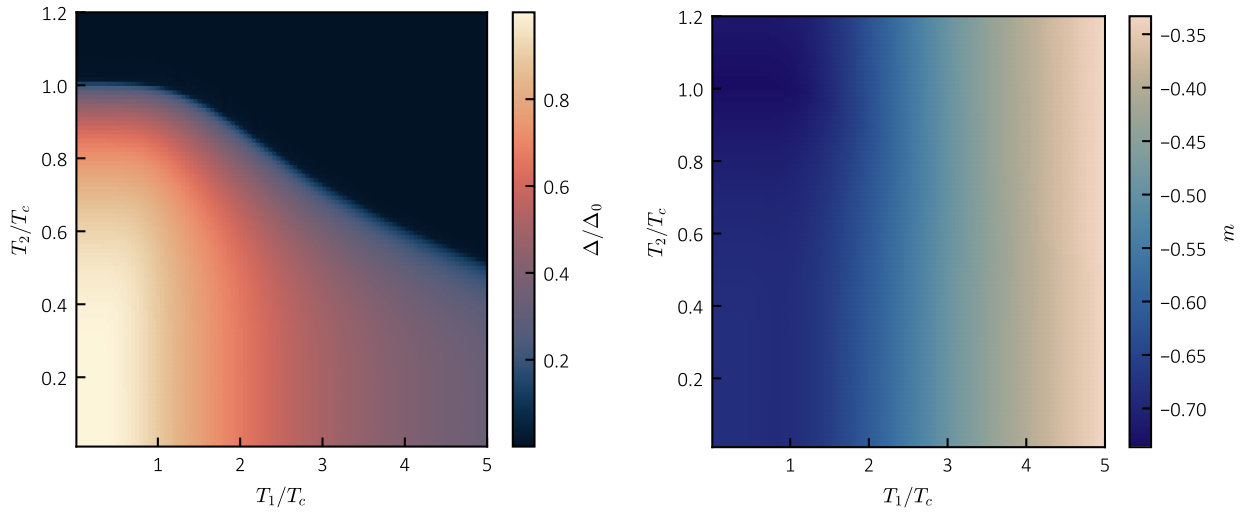
Slika 22 | Disperziji sta: $c = 10.0, \omega_0^{(1)} = 3, \omega_0^{(2)} = 0.01$. Grafi so v točki $T_1 = 6.0T_c$ in $T_2 = 1.3T_c$, kjer preidemo v BCS režim.



Slika 23 | Disperziji sta: $c = 10.0, \omega_0^{(1)} = 3, \omega_0^{(2)} = 0.01$. Grafi so v točki $T_1 = 6.0T_c$ in $T_2 = 1.3T_c$, kjer preidemo v BCS režim.

4.3 Primer NESS za BCS

Primer NESS v BCS režimu: $G = 2, t_a = -t_b = 1$ in $V = 4.4$, kjer je $\Delta_0 = 1.0$. Disperziji sta: $c = 10.0$, $\omega_0^{(1)} = 3.0$, $\omega_0^{(2)} = 0.01$. V sklopitvi je $d = 2$ in $\gamma = 0.15$ in $N_k = 50$.



Slika 24 | Disperziji sta: $c = 10.0$, $\omega_0^{(1)} = 3$, $\omega_0^{(2)} = 0.01$, BCS režim. Obe s fotonskim verteksom.

5 Pretvorbe enot

Do zdaj smo uporabljali $\hbar = k_B = 1$ in delali v enotah $a_0 = 1$ ter $t_a = 1$. Za pretvorbe si moramo izbrati konkretne vrednosti za $a_0 = 5$ Å in $t_a = 0.1\,\mathrm{eV}$. Za brezdimenzijske količine bom uporabljal kačo, le v tem odseku. Od tod so naravne pretvorbe za hitrosti:

$$v_0 = \frac{t_a a_0}{\hbar} \simeq 76\,\mathrm{km/s}\,. \tag{277}$$

Za izbrane vrednosti parametrov je potem hitrost svetlobe $\tilde{c} = c/v_0 \simeq 3950$. Za hitrost zvoka $\tilde{c}_s = 10\,\mathrm{km/s}/v_0 \simeq 0.13$. Pretvorba za količine povezane s časom je

$$\tau_0 = \frac{\hbar}{t_a} \simeq 6.6\,\mathrm{fs}\,. \tag{278}$$

Osnovno frekvenco izberemo, da je na vleikostni skali reže, torej nekaj 100 kelvinov, oziroma $\tilde{\omega}_0 \sim 0.1 \implies \omega_0 \simeq 1.5 \cdot 10^{13}\,\mathrm{s}^{-1}$. Dolžina resonatorja je v tem primeru $\tilde{L} = L/a_0 = \pi\tilde{c}/\tilde{\omega}_0 \simeq 1.2 \cdot 10^5 \sim 62\,\mu\mathrm{m}$. Za fotonsko sklopitveno konstanto velja:

$$\tilde{g} = \frac{g}{t_a^2} = \frac{e^2\hbar}{2\mathcal{V}\varepsilon_0\omega_{\mathbf{q}}}\left(\frac{t_a a_0}{\hbar}\right)^2 \frac{1}{t_a^2} = 2\pi\alpha\frac{\tilde{c}}{\tilde{\omega}_{\mathbf{q}}}\frac{a_0^3}{\mathcal{V}} = 2\alpha\frac{\tilde{\omega}_0}{\tilde{\omega}_{\mathbf{q}}}\frac{1}{N_k^2}\,. \tag{279}$$

Tu sem za volumen vzel $\mathcal{V} = N_k^2 a_0^3 \tilde{L}$. Uteži $1/N_k^2$ so že vštete v vsote in povprečja po momentih, zato je v končnem izrazu ne upoštevamo. V kodi je vrednost za amplitudo

$$\tilde{A}_{\text{fot.}} = \tilde{g}\tilde{\omega}_{\mathbf{q}} = 4\alpha\tilde{\omega}_0 \sim \frac{4}{137}\cdot 0.1 \simeq 0.00292\,. \tag{280}$$

Drugi red je sorazmeren z $\tilde{A}^2 \ll \tilde{A}$. Za fonone velja

$$\tilde{g} = \frac{g}{t_a^2} = \frac{\hbar}{2MN_k^2\omega_{\mathbf{q}}} = \frac{1}{2\tilde{M}}\frac{1}{N_k^2}\frac{1}{\tilde{\omega}_{\mathbf{q}}} \tag{281}$$

kjer je brezdimenzijska masa ionov $\tilde{M} = M\frac{a_0^2 t_a}{\hbar^2} \simeq 598M[\mathrm{u}]$. V kodi je torej amplituda za fononsko sklopitev za $M = 20$ u:

$$\tilde{A}_{\text{fon.}} = \frac{1}{2\tilde{M}} \approx 4.2 \cdot 10^{-5} \tag{282}$$

razmerje med amplitudama je

$$\frac{\tilde{g}_{\text{fon.}}}{\tilde{g}_{\text{fot.}}} = \frac{\sqrt{\tilde{\omega}_0^2 + q^2\tilde{c}^2}}{4\alpha\tilde{M}\tilde{\omega}_0|q|\tilde{c}_s}\,, \tag{283}$$

torej za $q \to 0$ fononi dominirajo v verteksu:

$$\frac{\tilde{g}_{\text{fon.}}}{\tilde{g}_{\text{fot.}}} = \frac{1}{q}\frac{1}{8\alpha\tilde{M}\tilde{c}_s} \tag{284}$$

za večje momente pa:

$$\frac{\tilde{g}_{\text{fon.}}}{\tilde{g}_{\text{fot.}}} = \frac{\tilde{c}}{\tilde{c}_s}\frac{1}{8\alpha\tilde{M}\tilde{\omega}_0} \tag{285}$$

realistični dipol znotraj celice znaša okoli $d = 0.1 - 1$ Å, torej $\tilde{d} \simeq 0.02 - 0.2$.

5.1 Limita $q \to 0$ za fotone

$$\frac{1}{(2\pi)^2}\int \mathrm{d}\mathbf{q}\,\Gamma_{\beta\mathbf{k}'\to\alpha\mathbf{k}} = \frac{2\alpha\omega_0}{\pi c^2}N(\Omega)\left(\sin^2(k_x)+\sin^2(k_y)\right)\left(4t\frac{\Delta}{\Omega}\right)^2\cdot$$
$$\left[\left(1+\omega_0^2\frac{\Omega^2-\gamma^2}{(\Omega^2+\gamma^2)^2}\right)\left(\arctan\left(\frac{\Omega-\omega_0}{\gamma}\right)+\arctan\left(\frac{\Omega+\omega_0}{\gamma}\right)\right)+\frac{\gamma\Omega\omega_0^2}{(\Omega^2+\gamma^2)^2}\ln\left(\frac{((\Omega-\omega_0)^2+\gamma^2)((\Omega+\omega_0)^2+\gamma^2)}{\omega_0^4}\right)\right] =,$$
$$=: N(\Omega)J(\Omega)\,, \qquad \Omega = 2\sqrt{\xi_{\mathbf{k}}^2+\Delta^2}\,. \tag{286}$$

Enačba za dinamiko je :

$$\partial_t n_{\alpha,\mathbf{k}} \simeq J(\Omega)\left[N(\Omega)n_{\beta,\mathbf{k}}(1-n_{\alpha,\mathbf{k}}) - (1+N(\Omega))n_{\alpha,\mathbf{k}}(1-n_{\beta,\mathbf{k}})\right] \tag{287}$$

5.2 Limita za fonone

$$\sum_{\mathbf{q}}\Gamma_{\gamma\mathbf{k}+\mathbf{q}\to\gamma\mathbf{k}} \simeq$$
$$\frac{(2t')^2}{2Mc_s^2}\int_0^{\omega_D} d\Omega\,\frac{1}{2\pi}\int_0^{2\pi} d\varphi\,\left(u_{\mathbf{k}+\mathbf{q}}u_{\mathbf{k}} - v_{\mathbf{k}+\mathbf{q}}v_{\mathbf{k}}\right)^2\left[\left(\sin(k_x+q_x)-\sin(k_x)\right)^2+\left(\sin(k_y+q_y)-\sin(k_y)\right)^2\right]\cdot$$
$$\cdot\left[(1+N(\Omega))+N(\Omega))\right]\delta(E_{\mathbf{k}+\mathbf{q}}-E_{\mathbf{k}}-\Omega)\,;$$
$$\mathbf{q}(\Omega,\varphi) = \frac{\Omega}{c_s}(\cos\varphi,\sin\varphi) \tag{288}$$

$$\left(u_{\mathbf{k}+\mathbf{q}}u_{\mathbf{k}} - v_{\mathbf{k}+\mathbf{q}}v_{\mathbf{k}}\right)^2 = \frac{1}{2}\left(1+\frac{\xi_{\mathbf{k}+\mathbf{q}}\xi_{\mathbf{k}}-\Delta^2}{\sqrt{\Delta^2+\xi_{\mathbf{k}+\mathbf{q}}^2}\sqrt{\Delta^2+\xi_{\mathbf{k}}^2}}\right) \tag{289}$$

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